

VOIP OVERVIEW

for Dual Education - TPS

by Tomas Stasik

T · Systems ·

WHAT IS VOIP

Voice over Internet Protocol (VoIP) is a technology that enables one to make and receive phone calls through the Internet instead of using the traditional analogy PSTN (Public Switched Telephone Network) lines.



FROM PSTN TO VOIP

The Public Switched Telephone Network, also known as **Plain Old Telephone Service (POTS)**

The PSTN relies on **circuit switching**.

Problem?

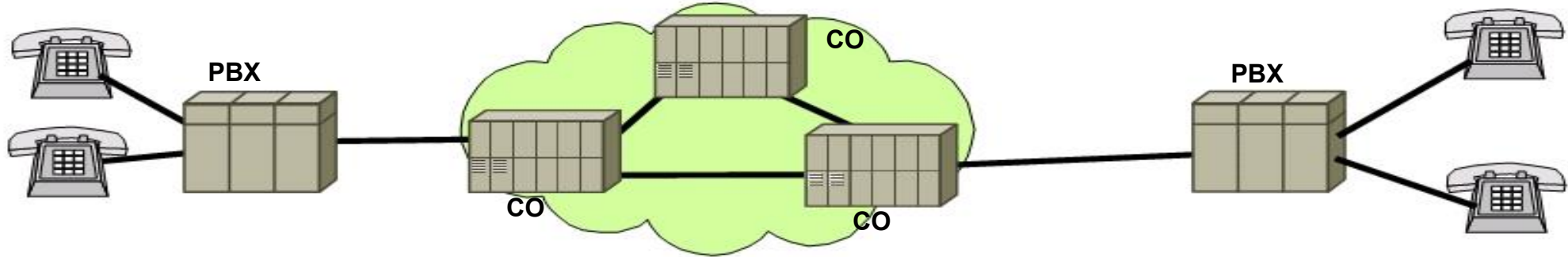
How to share line...



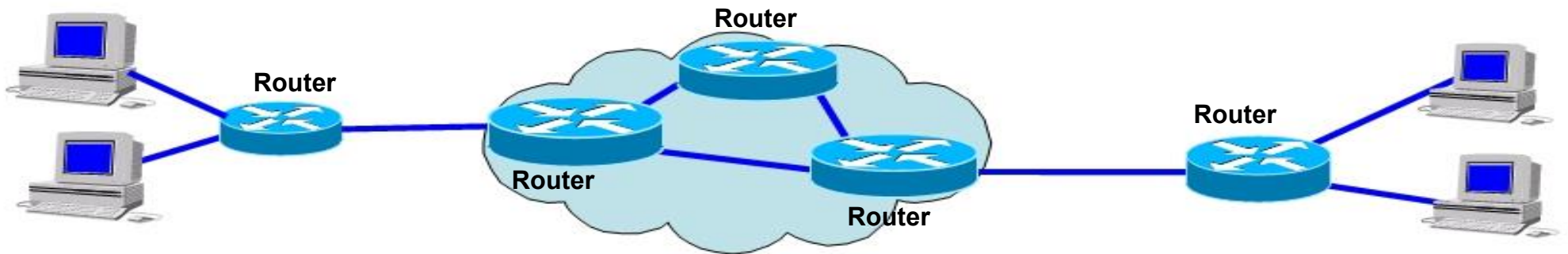
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FROM PSTN TO VOIP

Circuit Switched Networks (Voice)

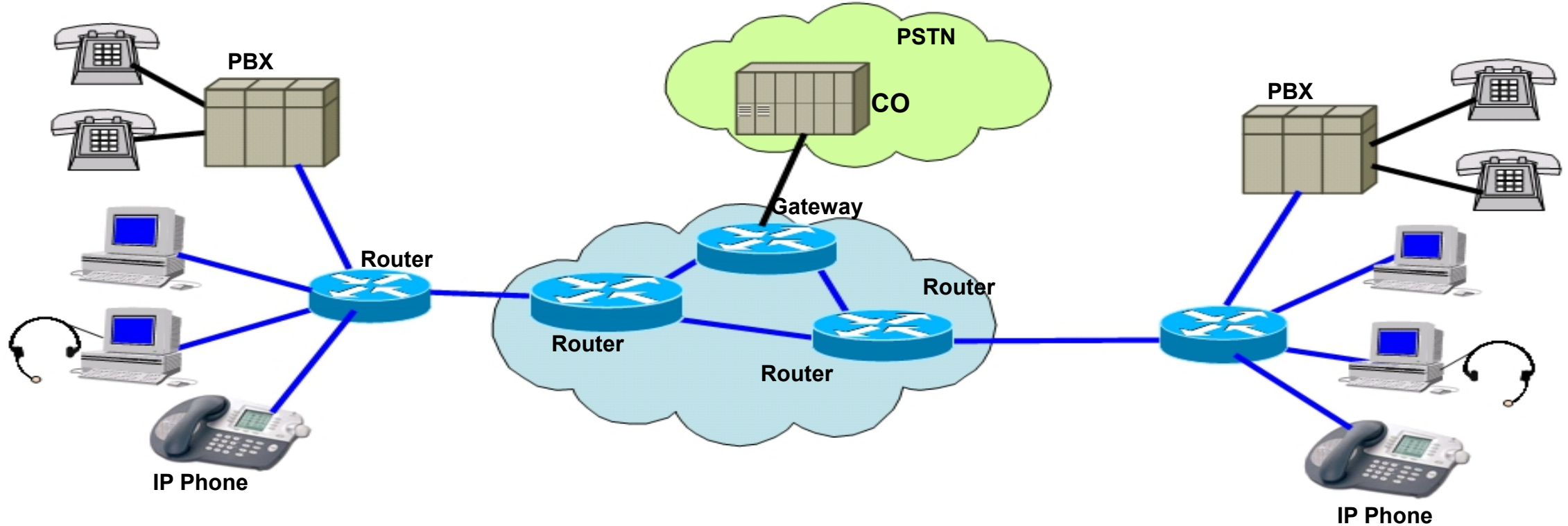


Packet Switched Networks (Data)



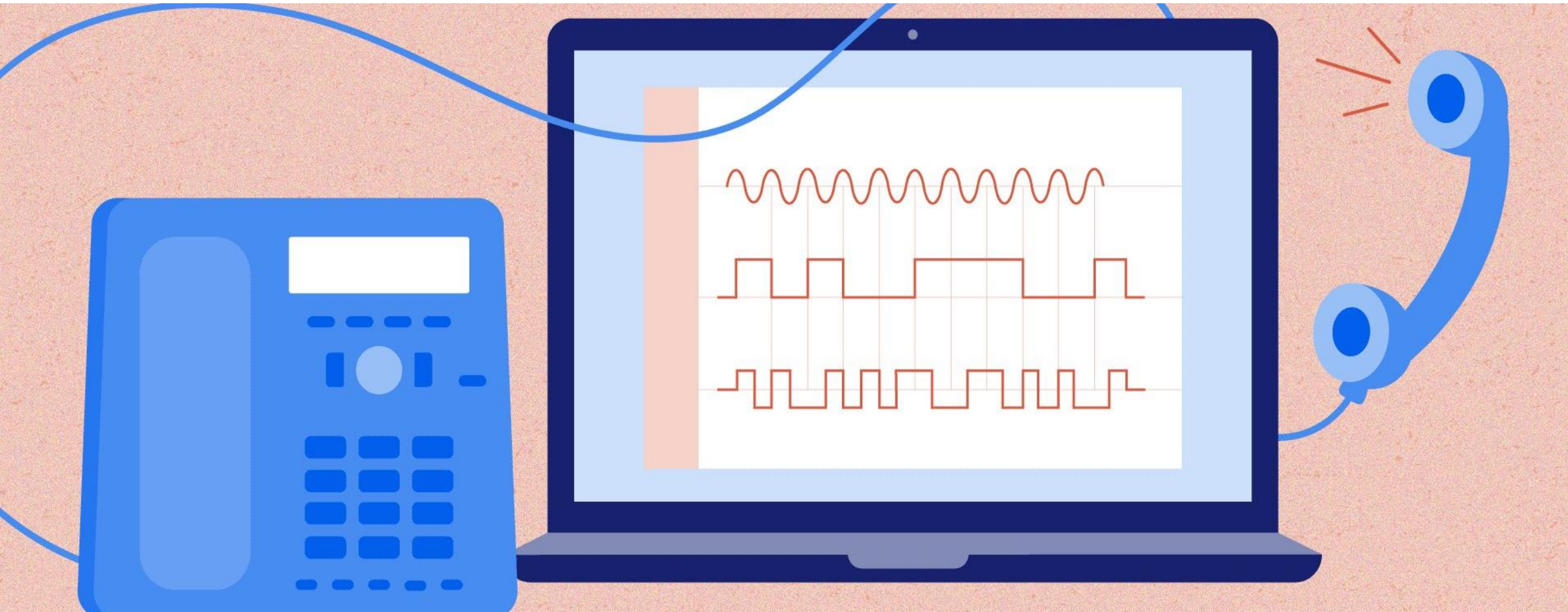
- Separated networks
- Separated applications/services

FROM PSTN TO VOIP



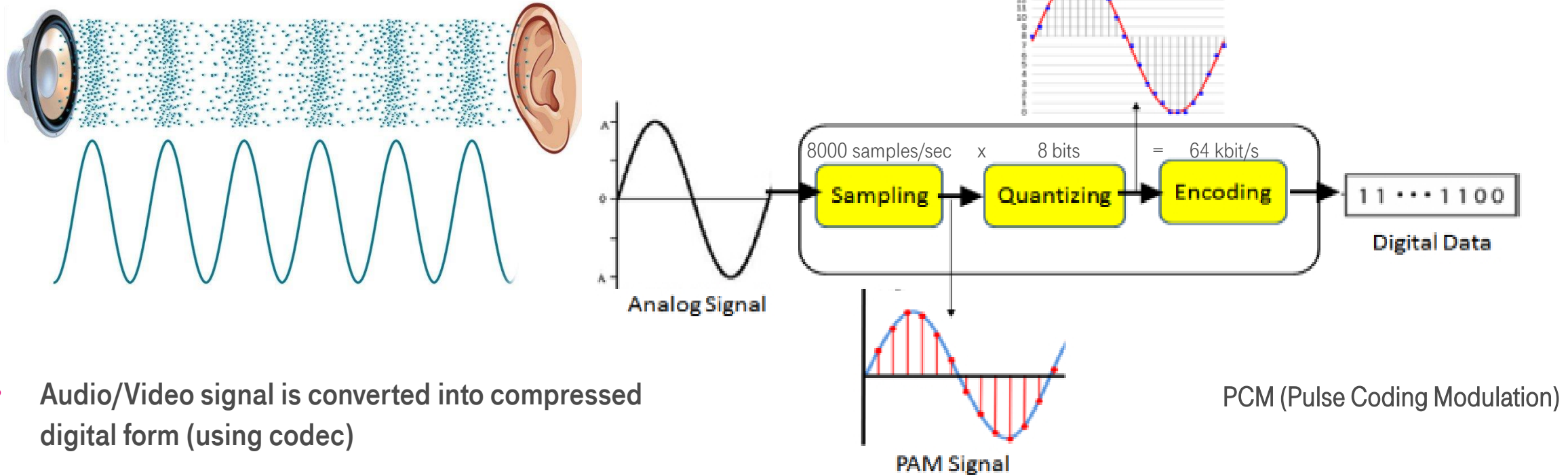
- Converged network
- Separated or integrated applications

HOW VOICE DATA ARE CREATED



VOICE DIGITIZATION -> CODEC (CODING-DECODING)

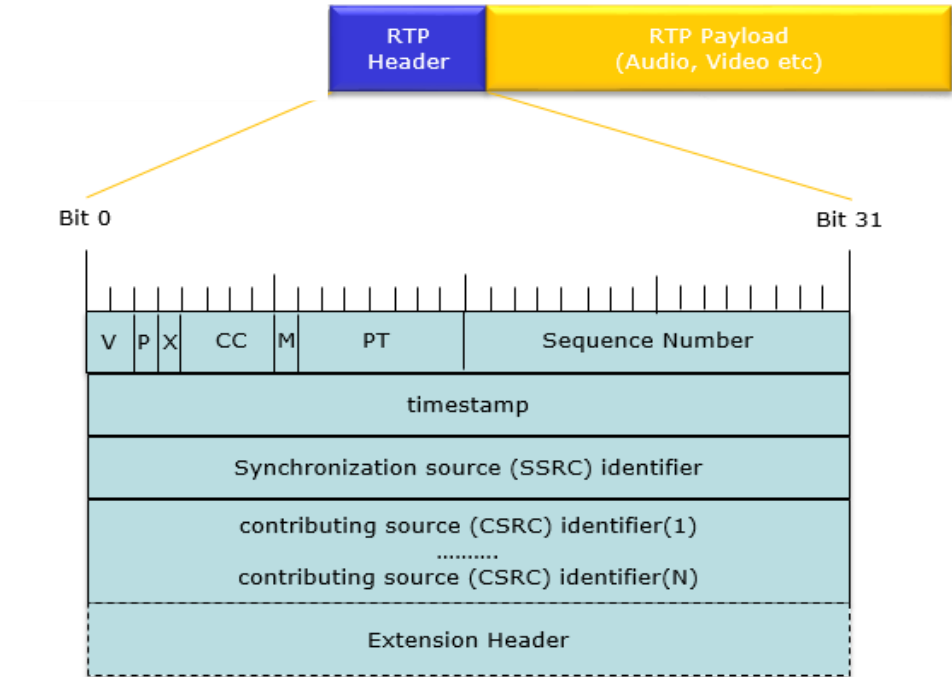
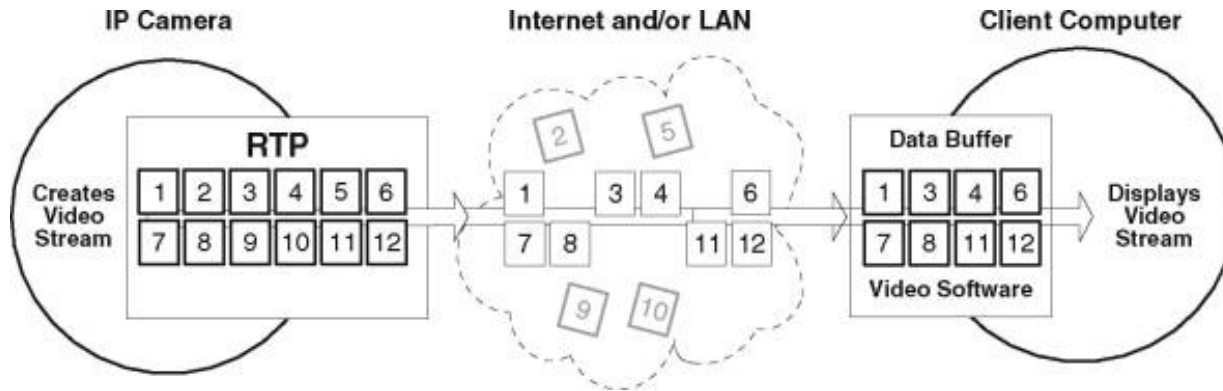
- sound is a pressure wave created by a vibrating object



- Audio/Video signal is converted into compressed digital form (using codec)
- Mainly used is PCM (G.711) – analog audio signal is encoded to binary code with data speed **64kbps**

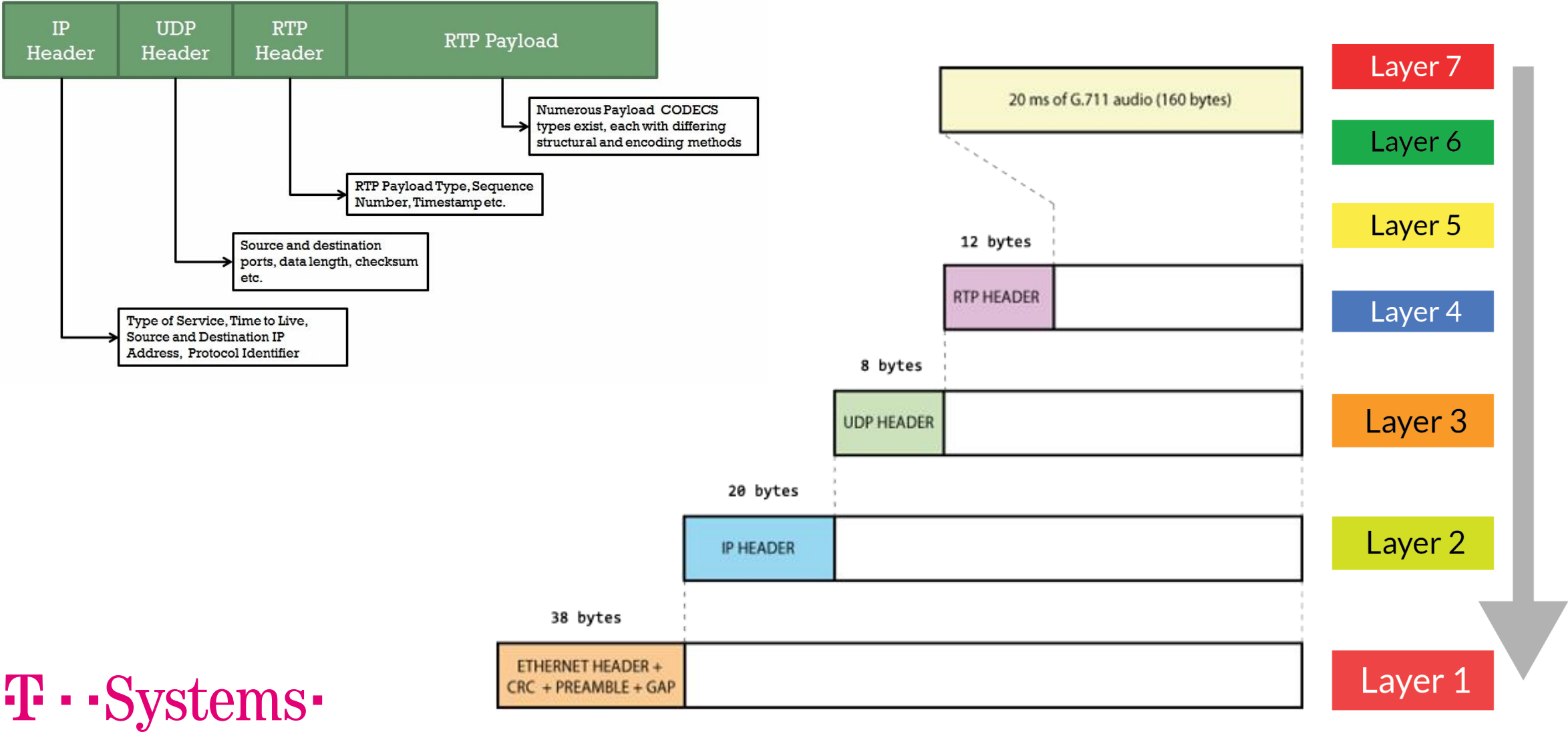
REAL-TIME TRANSPORT PROTOCOL

- Audio signal (payload) is encapsulated into RTP protocol on application layer
- Audio communication is End-to-End
- Packets may arrive out of order, RTP fix this with timestamp and Sequence No.



ENCAPSULATION

Client Side



BANDWIDTH CALCULATION FOR G.711

Let's send 1 audio frame (20ms audio payload):

Bytes transmitted every 20ms:

$$160 + 12 + 8 + 20 + 38 = 238 \text{ bytes}$$

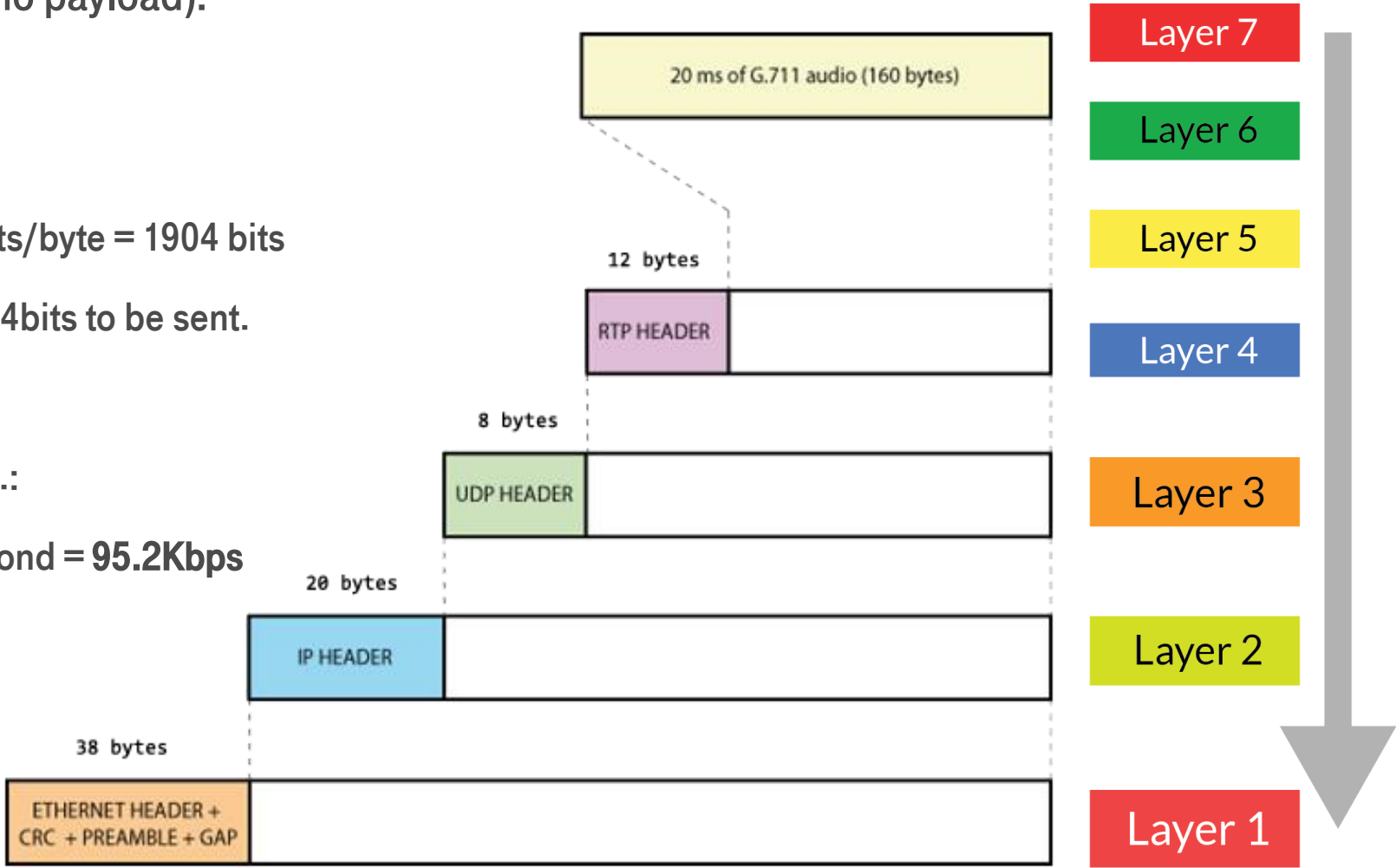
transformed to bits: $238 \text{ bytes} * 8 \text{ bits/byte} = 1904 \text{ bits}$

so simple 20ms audio frame means 1904bits to be sent.

1 second has 50 such 20ms frames, e.g.:

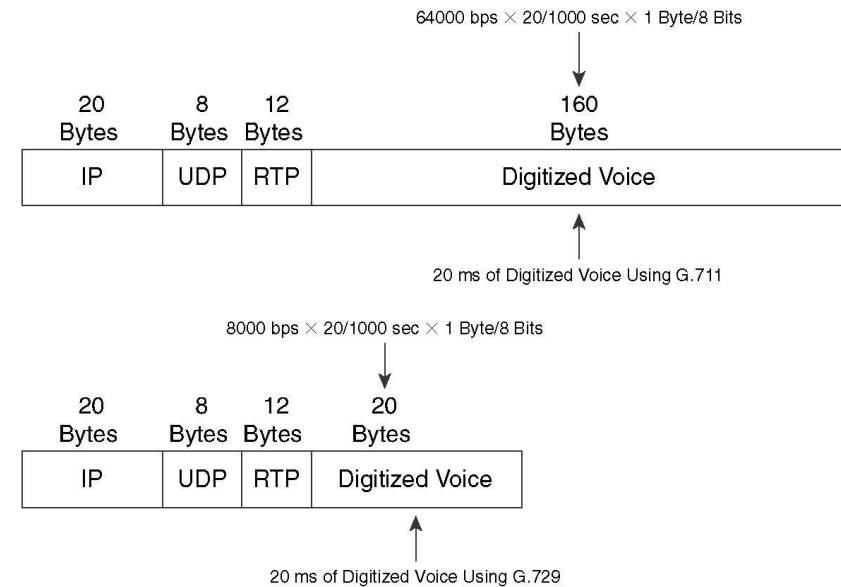
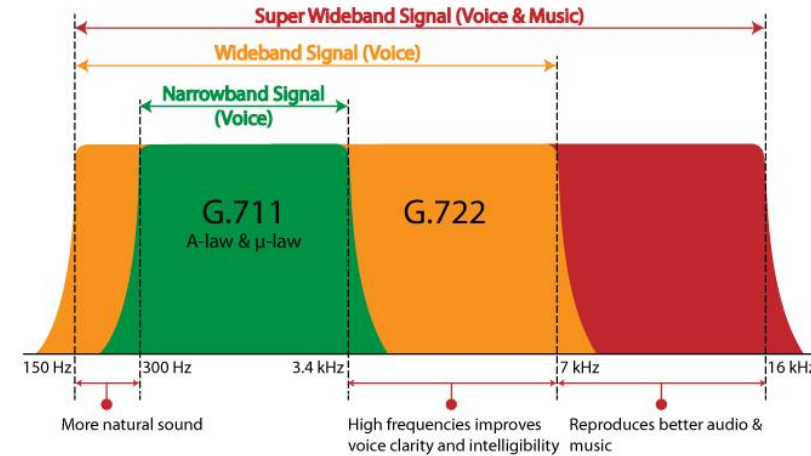
$$1904 \text{ bits} * 50 \text{ frames} = 95,200 \text{ bits/second} = 95.2 \text{ Kbps}$$

Client Side



MOSTLY USED CODECS

- There are existing various codec – each is suitable for different purpose or application
- G.711 (PCM) is “default” for all
 - it uses PCM method -> there are 2 modifications “uLaw” (PCMU) used in USA and “aLaw” (PCMA) used in EU
 - each VoIP device must support
 - 64kbits
 - typically used for internal LAN network
 - Good audio quality of speech
- G.729
 - often codec used for external (WAN) connection
 - There are several modifications G.729a, G729b, G729ab
 - 8kbits
 - Low audio quality of speech



COMPARING CODEC BW

Table for the actual bandwidth for common codecs (using Ethernet):

Codec	BW codec	BW actual (Ethernet)
G.711	64 Kbps	95.2 Kbps
G.726*	32 Kbps	63.2 Kbps
iLBC*	15.2 Kbps	46.4 Kbps
GSM	13 Kbps	43.7 Kbps
G.729	8 Kbps	39.2 Kbps

One of the method how to save BW is Voice Activity Detection (**VAD**) - is a technique used in speech processing in which the presence or absence of human speech is detected. It can avoid unnecessary coding/transmission of silence packets in VoIP.

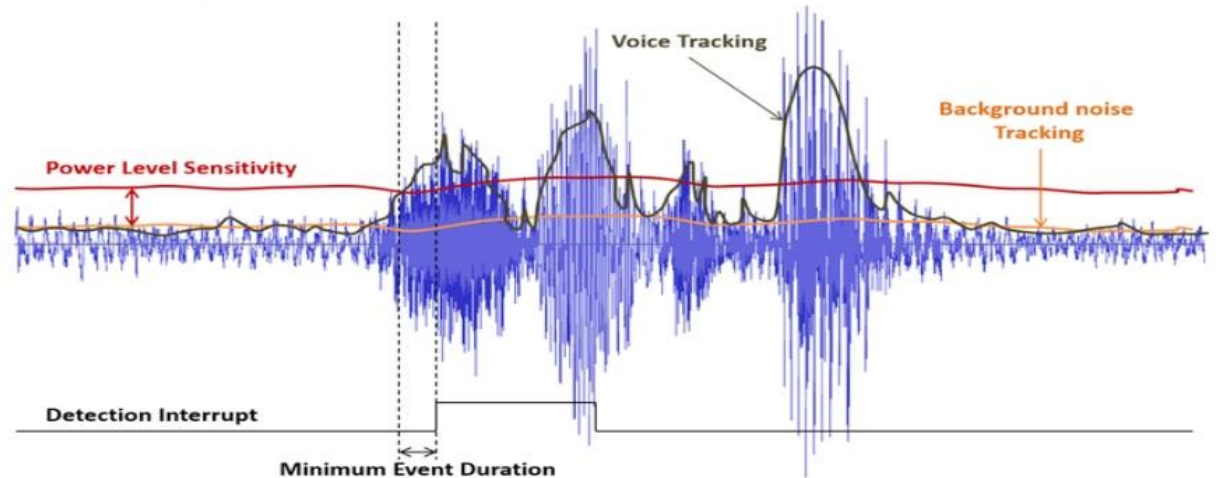
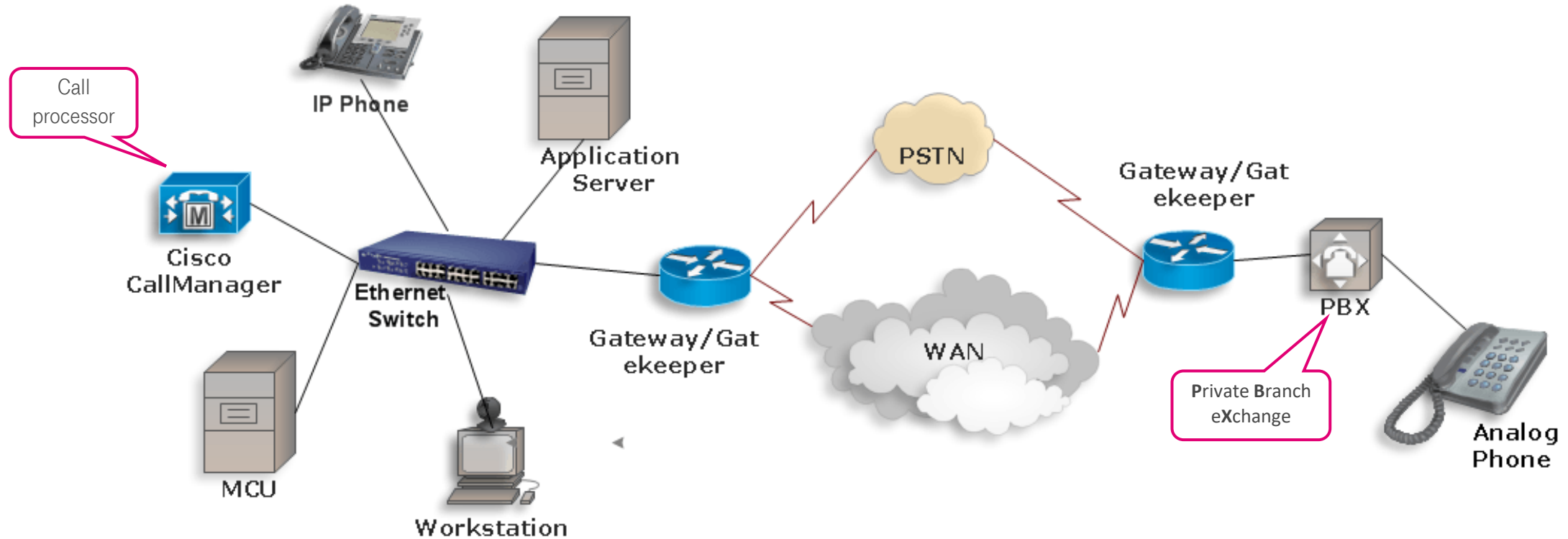


Figure 3: Example of wake-up signal

VOICE COMPONENTS



COMPONENTS OF VOIP NETWORK



ENDPOINTS

- IP (Hardware) Phone; Softphone (Webex Teams, etc.); Telepresence device (WebexBoard, etc.)

At your desk

- Headsets: 300 / 500 / 700 / 900 Series
- USB camera: Webex Desk Camera
- IP phones: 6800 / 7800 / 8800 / 9800 Series
- All-in-one premium collaboration and co-creation: Webex Desk Series - Hub / Mini / Desk / Desk Pro

Meeting spaces

- IP conferencing: 7832, 8832
- Video-first kits: Webex Room Kit Series
- Video-first integrated systems: Webex Room Series
- Immersive boardroom: Webex Room Panorama Series
- Team collaboration and co-creation: Webex Board Series

Single platform advantage

Mobile

- Ruggedized WiFi Phones: DBS-110, DBS-210, Cisco 840/860 (coming soon)
- ATAs: Cisco 191/192, AudioCodes MP-124E/MP-1288, VG400, VG420
- Consistent user experiences: Webex App
- Single pane of glass management and analytics: Control Hub

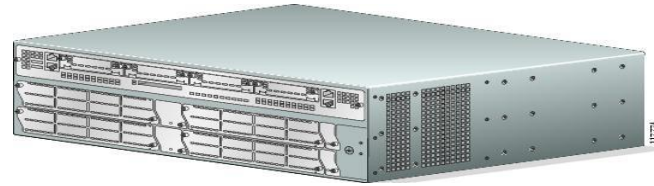
GATEWAYS – VOIP EXTERNAL INTERFACES

“Analog” VGW

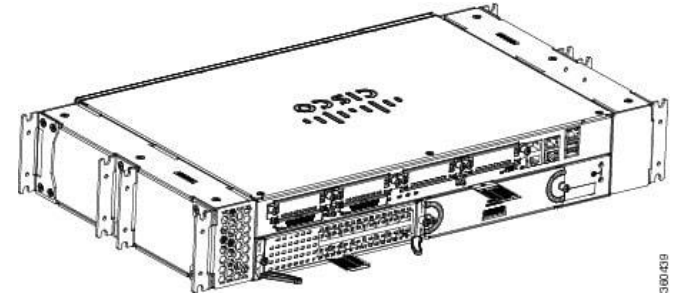
VG202, VG204, VG224, etc.



“Digital” VGW / PSTN GW



SIP Trunk - CUBE



CALL PROCESSOR

Main call-routing component which is processing all calls

- Analog/Digital PBX
- IP PBX (3CX, Asterisk, Cisco UCM, etc.)

Cisco Unified Communications Manager (CUCM) is the main basic call-routing component - it behaves as PBX.

Can be also running on IOS -> CUCM Express



HOW VOICE COMPONENTS COMMUNICATES

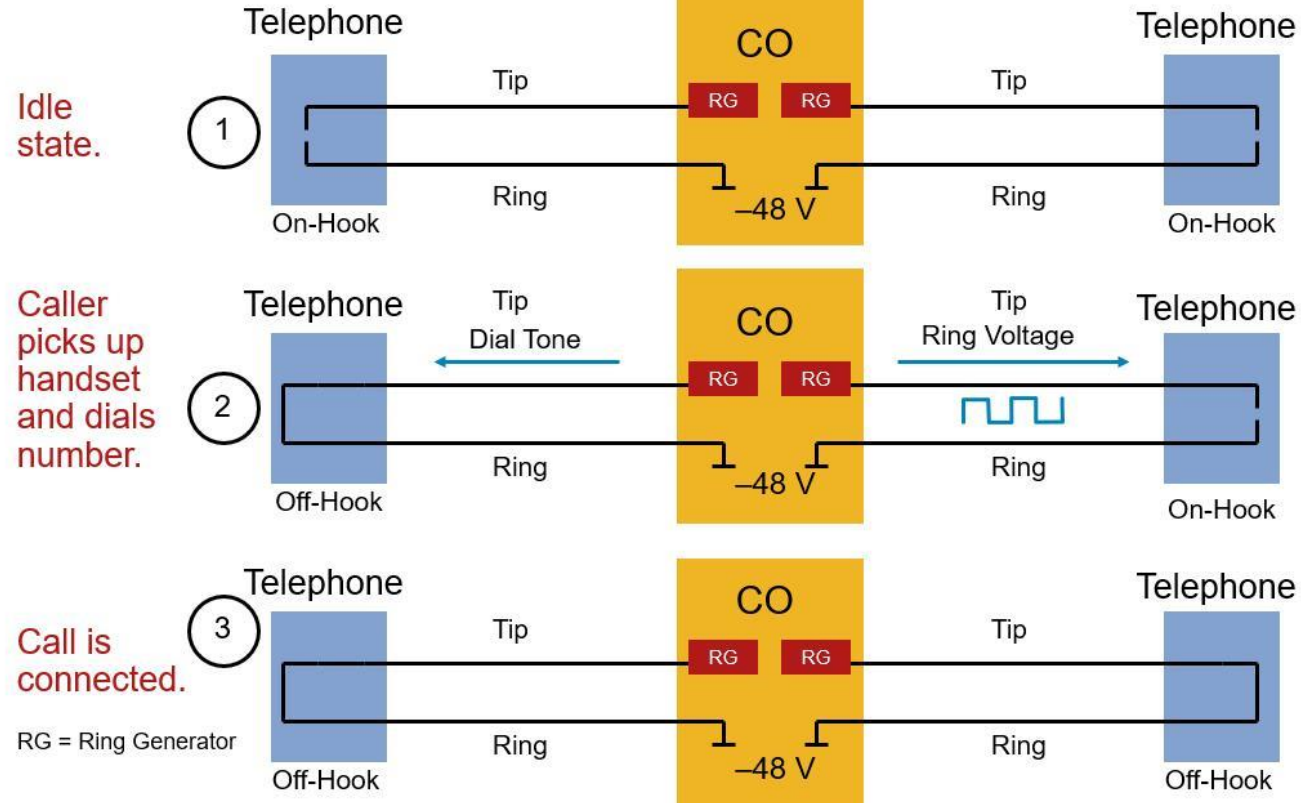


OLD SIGNALING

When device wants to communicate, it has to send signaling message.

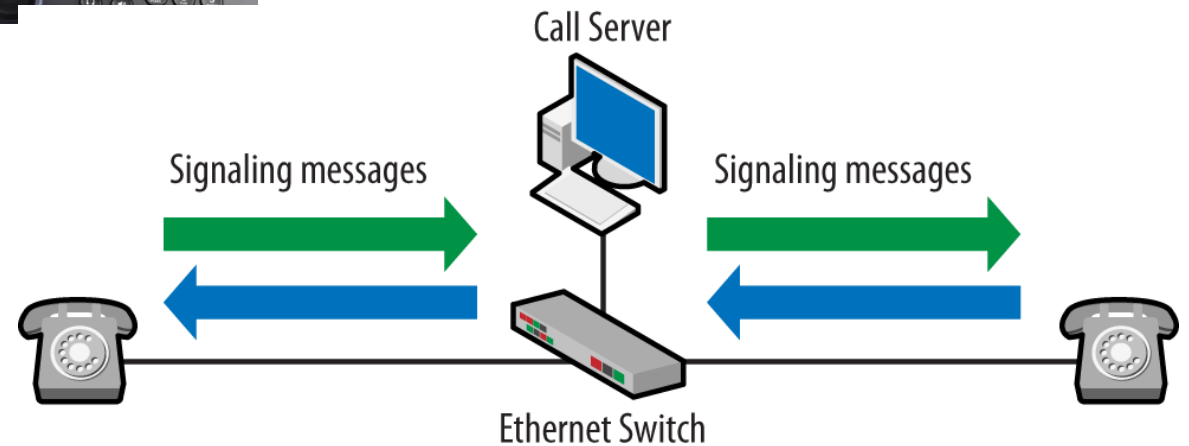
In analog world was signaling performed by loop start or ground start.

VoIP signaling is used for establishing, changing or ending sessions (audio call) in an IP network.



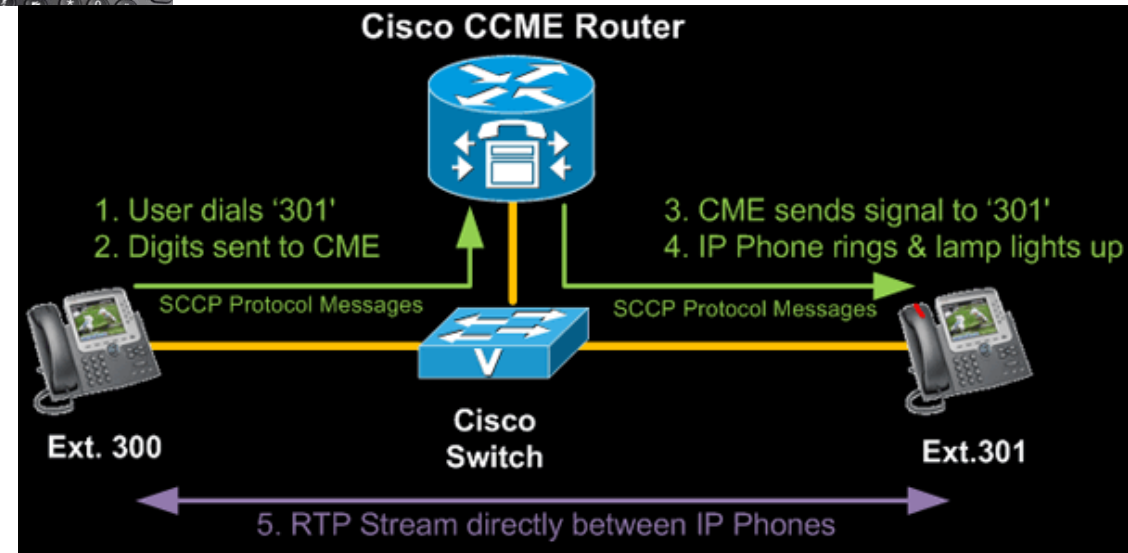
HOW DOES VOIP SIGNALING WORK

- Each VoIP device has to communicate
- Signaling messages in VoIP are sent via signaling protocols



HOW DOES VOIP SIGNALING WORK

- Each VoIP device has to communicate
- Signaling messages in VoIP are sent via signaling protocols
- Signaling is separate communication!
- Signaling messages are generated on App layer



SIGNALING PROTOCOLS

SIP = Session Initiation Protocol (port 5060, 5061)

H.323 = consists of various protocols (port 1720)

MGCP = Media Gateway Control Protocol (port 2427, 2727)

SCCP = Skinny Client Control Protocol (port 2000)

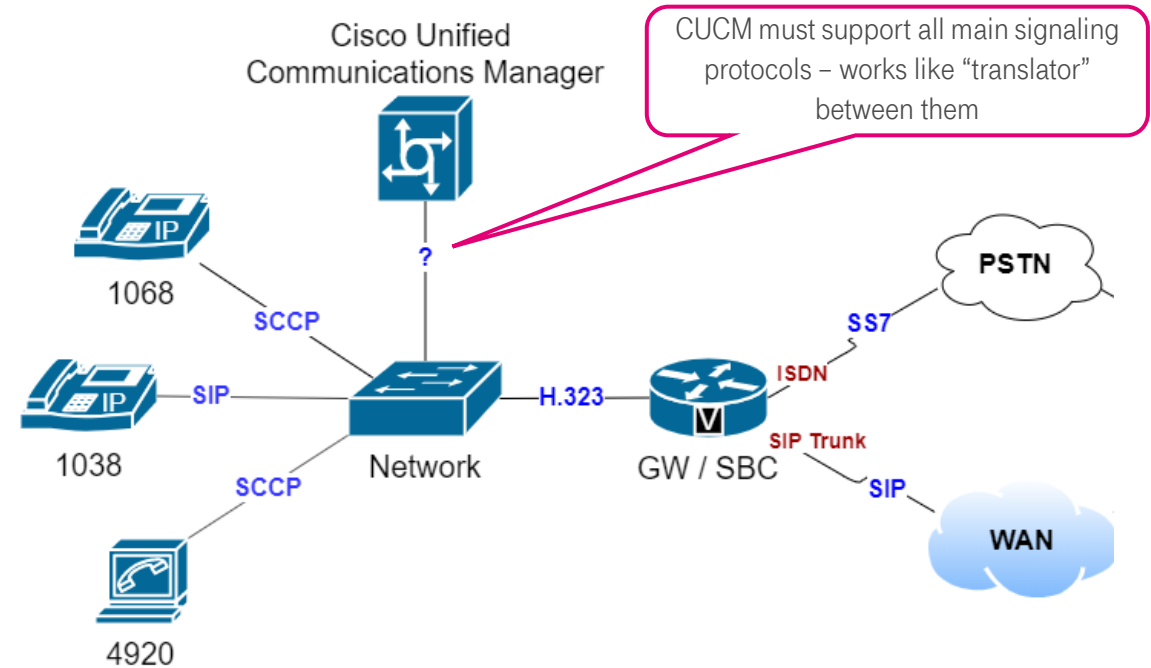
SS7 = signaling within PSTN network

(SS7 interoperability within IP networks provides SIGTRAN)

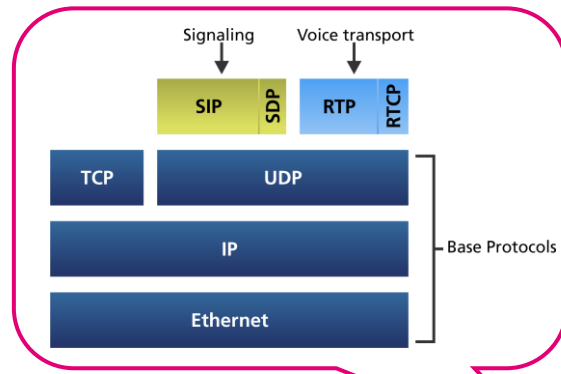
RTP is not signaling protocol, but transport!

Port range is per default from 16384 to 32767.

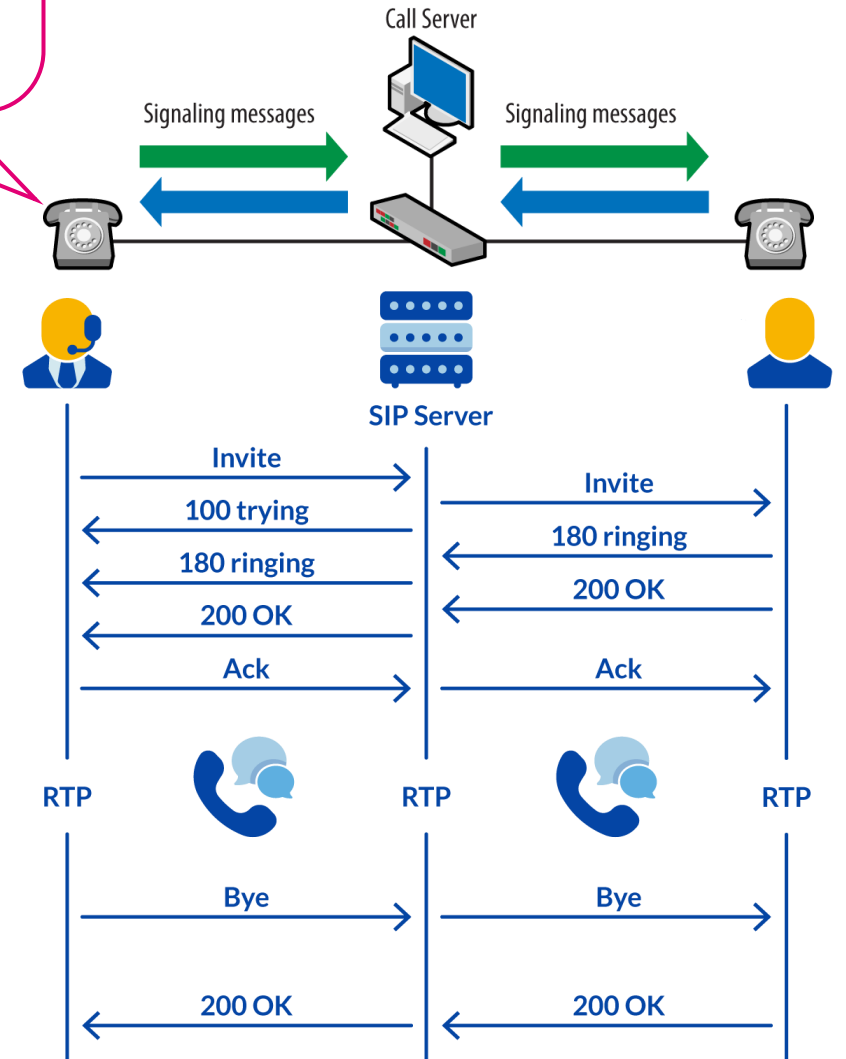
While RTP is delivered to an even-numbered port,
the corresponding RTCP must be the next higher odd-numbered port



SIP COMMUNICATION

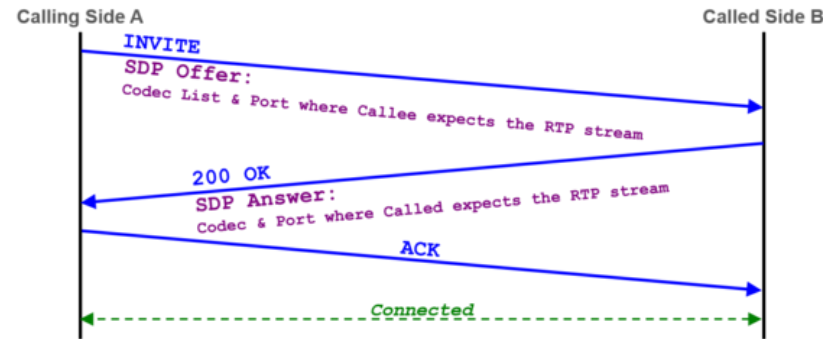


- SIP message has concept similar to HTTP message
- client-server protocol
- SIP dialog = SIP offer (INVITE, BYE, ...) + SIP response is in code (200 OK, 404 not found, ...)
- Default ports 5060 or 5061 (encrypted SIP)
- Nowadays is SIP most used signaling protocol.



SIP MESSAGE EXAMPLE

- SIP has header and body :
 - message body can contain SDP (Session Description Protocol)
 - SDP contains call parameters (media type, IP, codecs, etc.)



```
INVITE sip:sip2@sip.com SIP/2.0

Via: SIP/2.0/UDP 10..0.100;branch=z9hG4bK1234
Max-Forwards: 70
To: <sip:sip1@sip.com>
From: <sip:sip2@sip.com>;tag=123456
Call-ID: 3d45f59a12b54
CSeq: 1 INVITE
Contact: sip:10.2.0.100:5060
Content-Type: application/sdp
Content-Length: 100

v=0
o=- 5894032 5894032 IN IP4 10.2.0.100
S=SDP Media Session
... The rest is omitted.
```

Starting Line

Header Field

Separator

Message Body

IP address where the audio RTP stream is expected

UDP port where the audio RTP stream is expected

List of offered or accepted codec for voice

Voice codec (Top: first priority)

Directive to send and/or receive RTP

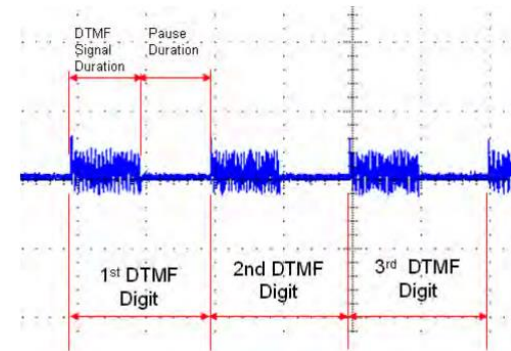
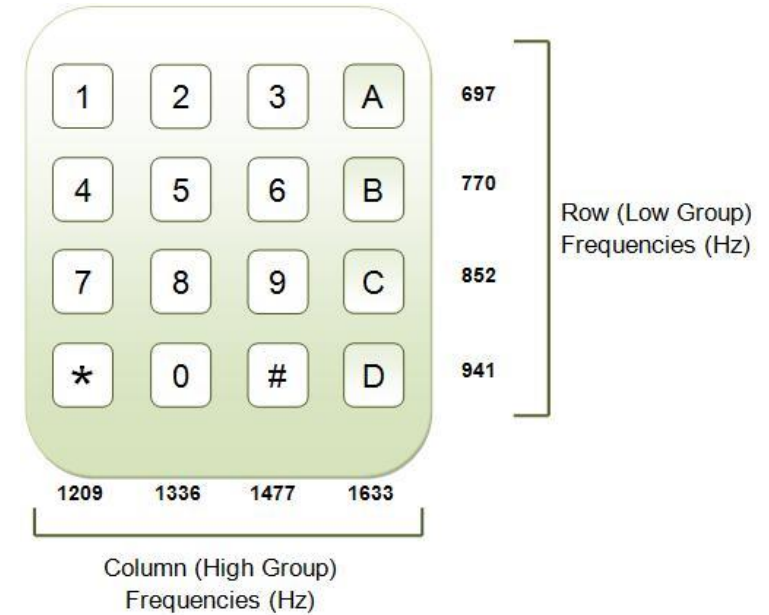
```
Session Description Protocol
  Session Description Protocol Version (v): 0
  Owner/Creator, Session Id (o): MxSIP 0 14205 IN IP4 62.12.246.233
  Session Name (s): SIP Call
  Connection Information (c): IN IP4 62.12.246.233
  Time Description, active time (t): 0 0
  Media Description, name and address (m): audio 5174 RTP/AVP 2 18 0 125 101
    Media type: audio
    Media Port: 5174
    Media Proto: RTP/AVP
    Media Format: ITU-T G.721
    Media Format: ITU-T G.729
    Media Format: ITU-T G.711 PCMU
    Media Format: 125
    Media Format: 101
    Media Attribute (a): rtptime:2 G726-32/8000
    Media Attribute (a): rtptime:18 G729/8000
    Media Attribute (a): rtptime:0 PCMU/8000
    Media Attribute (a): rtptime:125 X-CLEAR-CHANNEL/8000
    Media Attribute (a): rtptime:101 telephone-event/8000
    Media Attribute (a): fmp:18 annexb=no
    Media Attribute (a): fmp:101 0-16
    Media Attribute (a): sendrecv
```

Bit transparent data transfer

DTMF transfer RFC2833 (with PT101)

DTMF (DUAL TONE MULTI-FREQUENCY)

- sounds or tones generated by a telephone when numbers are pressed
 - For example, when you press the “5” key on your phone’s keypad a combined signal tone of 1336 Hz and 770 Hz is sent
- DTMF can be transmitted within audio (In Band) or as a separate SIP message (Out of Band)
- Timing synchronization is critical



ADVANTAGES AND DISADVANTAGES (PSTN VS VOIP)

Flexibility

Low-cost

Reliability

Delay (latency) max 150ms

Jitter (max 30ms)

Packet loss (max 1%)

Power dependent

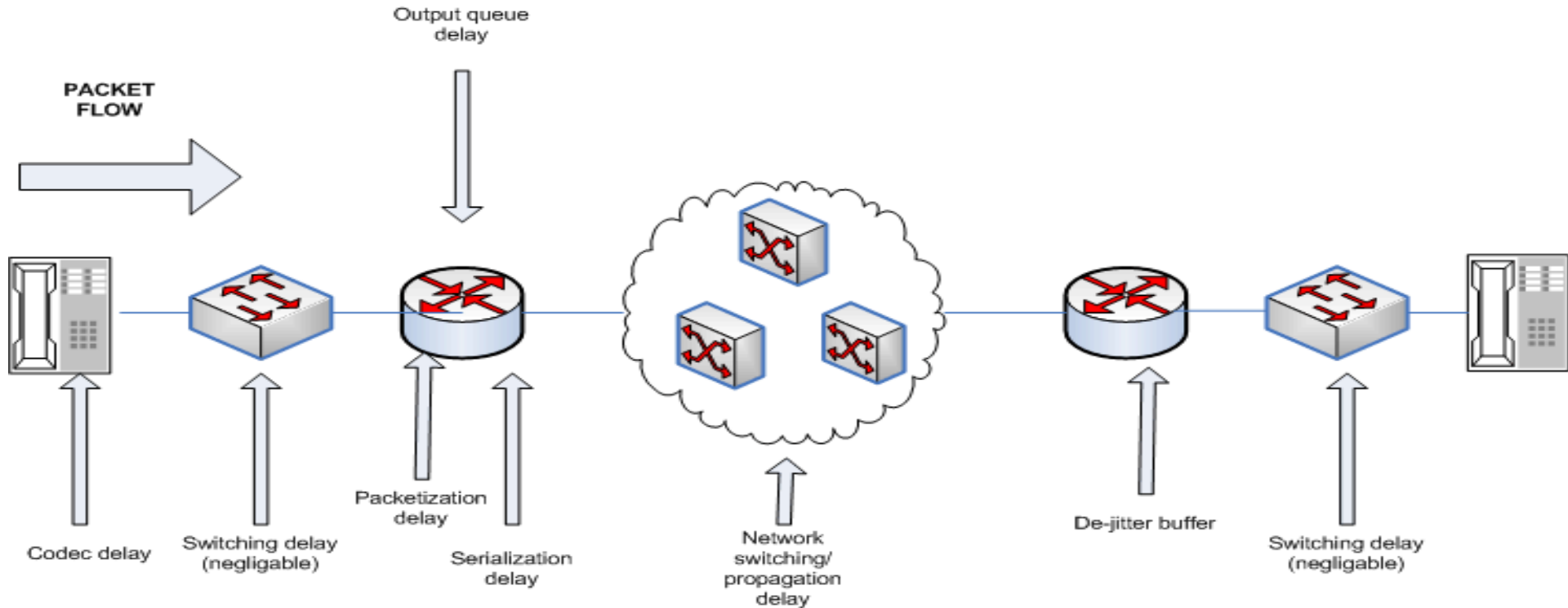
Emergency tracking

VOICE QUALITY CAN BE INFLUENCED BY MANY FACTORS LIKE BANDWIDTH, LATENCY, CODEC SUPPORT, THE TYPE OF ECHO CANCELLATION IN USE, WEATHER CONDITIONS

DELAY AND JITTER

Delay is caused when packets of data (voice) take more time than expected to reach their destination.

The delay is not always constant - variation in delay is called **jitter**. In VoIP, jitter is the absolute value of the difference between actual packet arrival time and the expected packet arrival time.



SOLUTION -> USING QOS (QUALITY OF SERVICE)

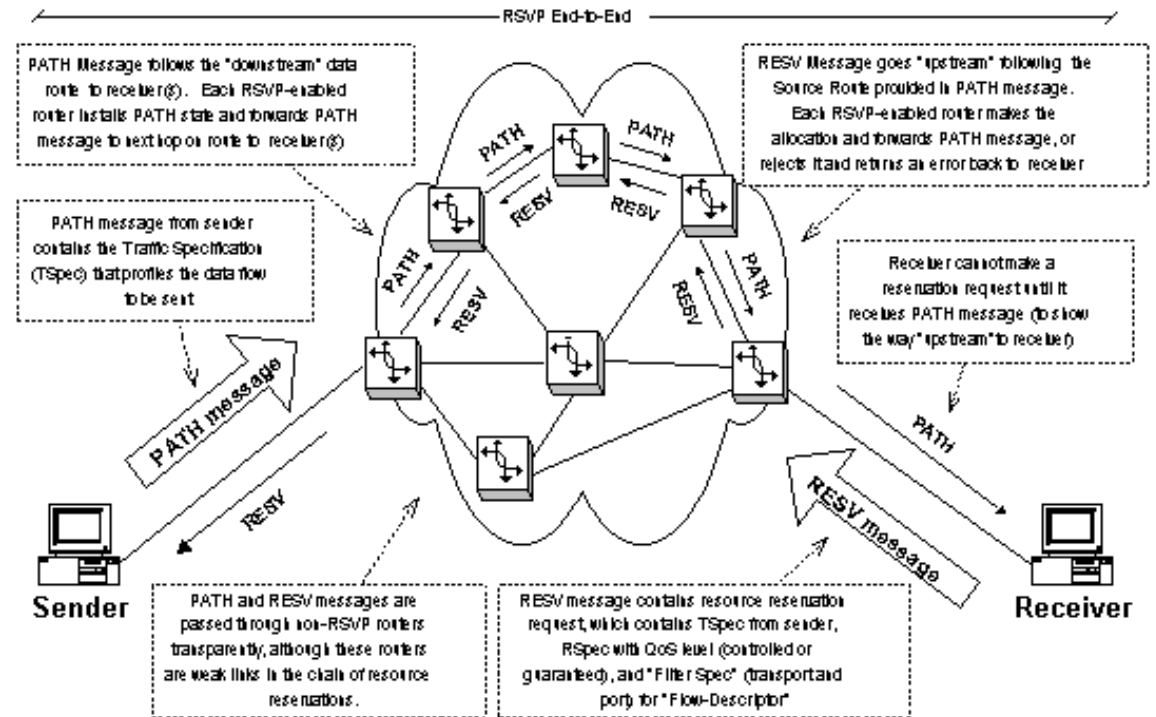
QoS Protocols:

- DiffServ: Differentiated Services IETF
- IP QoS: QoS in IP headers
- MPLS: Multiprotocol Label Switching IETF
- RSVP : Resource Reservation Protocol, IETF
- SBM

DiffServ: uses 6-bit differentiated services code point (DSCP) setting in IP packets or source and destination addresses

- Expedited Forwarding (EF) - Has a single codepoint (DiffServ value).
- Assured Forwarding (AF) - Has four classes and three drop-precedences within each class (so a total of twelve codepoints).

RSVP:



MANAGEMENT OF IP TELEPHONY



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CISCO UNIFIED COMMUNICATIONS MANAGER

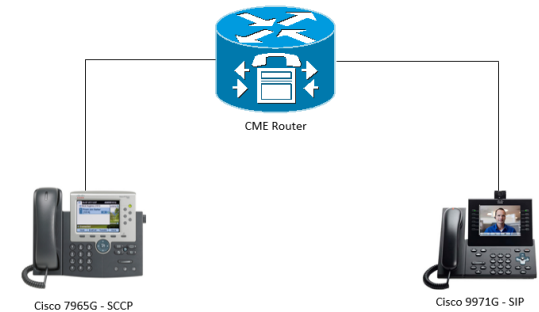
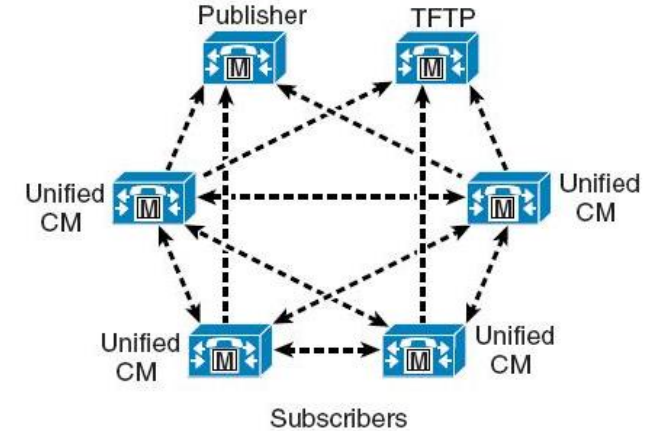
Cisco Unified Communications Manager (CUCM) is the main basic call-routing component - it behaves as PBX

CUCM supports clustering of servers to provide high availability and scalability. Database redundancy is provided by sharing a common database.

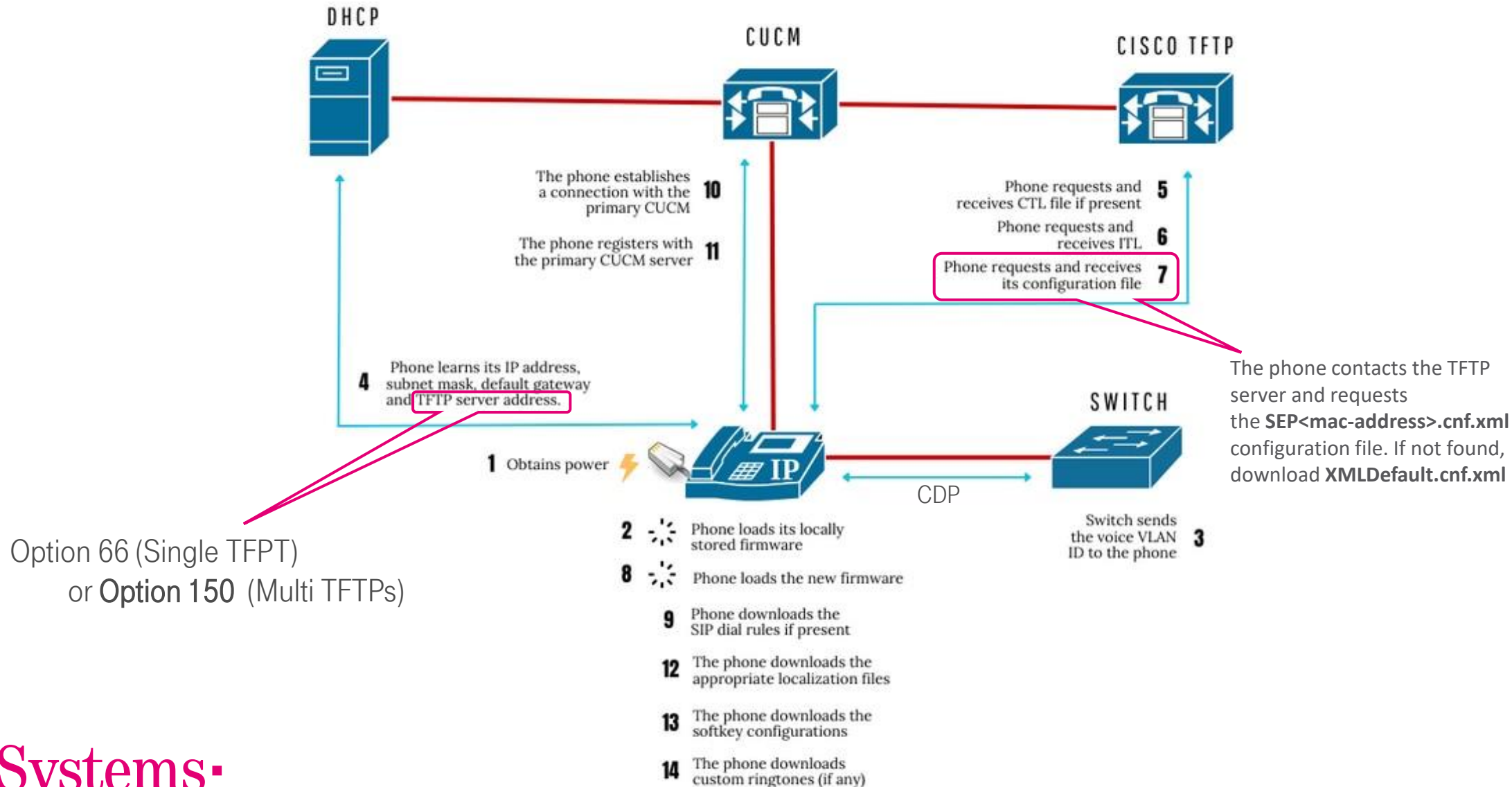
3 types of servers:

- Publisher (main DB server)
- Subscriber (communicating with endpoints)
- TFTP (storage of config files)

IOS version is **Call Manager Express** - runs within the local router instead of in its own node in the core network.



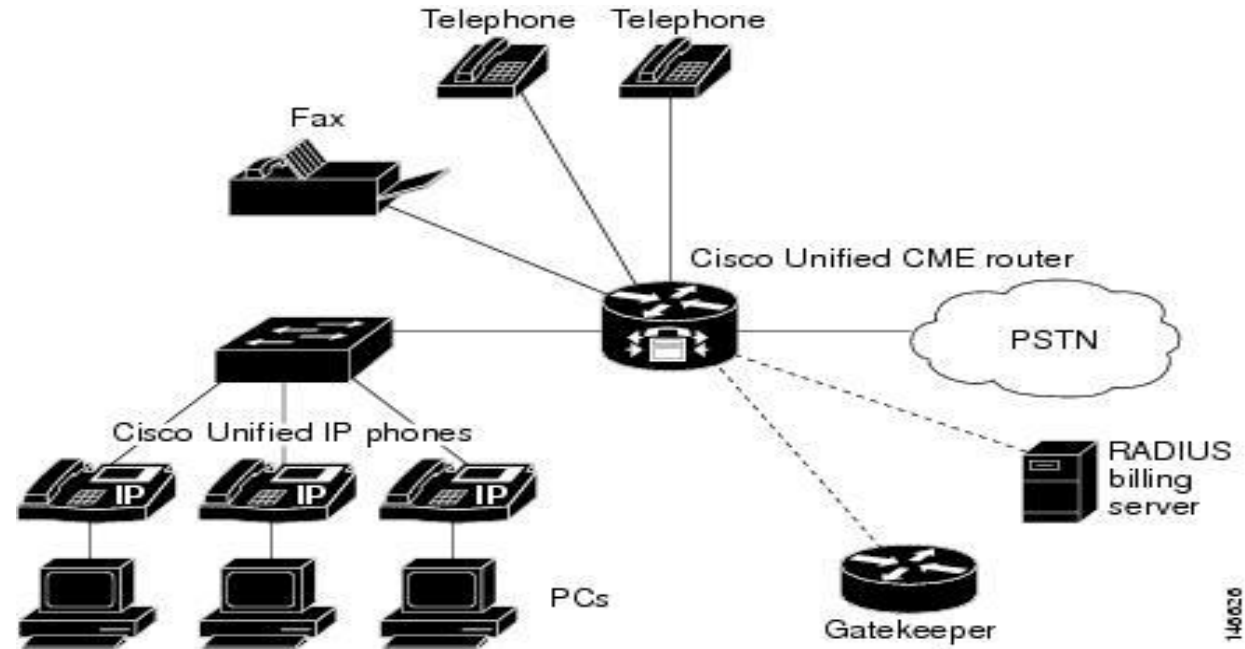
PHONE BOOT PROCESS (SIP PHONE)



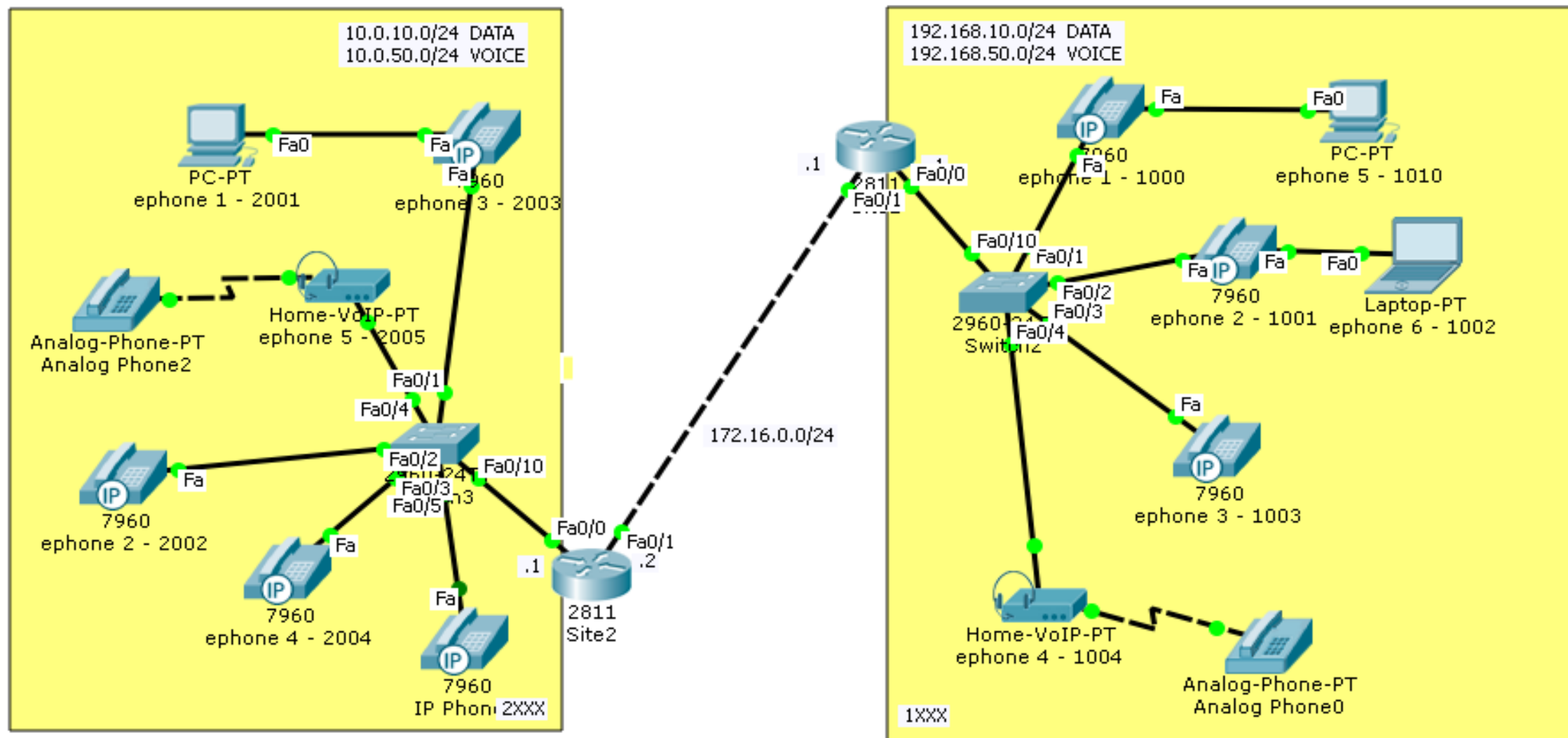
CISCO CALL MANAGER EXPRESS

Cisco Unified Communications Manager Express (formerly known as Cisco Unified CallManager Express) is a call-processing application in Cisco IOS software that allows small enterprise branch offices to deploy voice, data, and IP telephony on a single platform

is extremely flexible because it is modular. A Cisco Unified CME system consists of a router that serves as a gateway and one or more VLANs that connect IP phones and phone devices to the router.



LAB EXERCISE



STEPS

ROUTER:

Interfaces and subinterfaces (for VLAN)

DHCP - is needed to provide an IP address and the TFTP server location for each IP phone connected to the network.

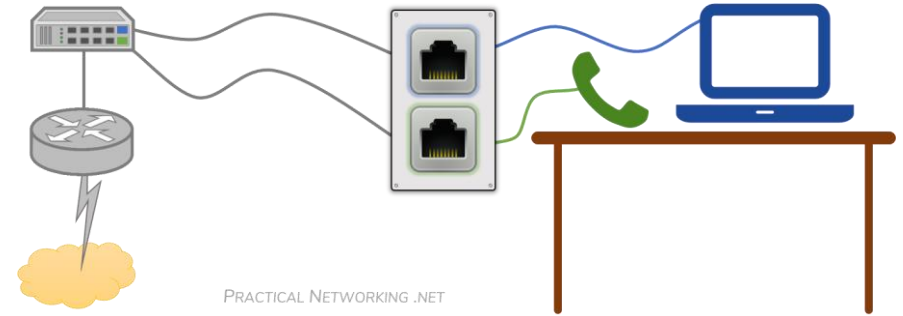
Routing

SWITCH:

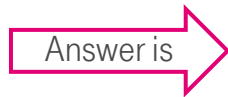
Configure interfaces and VLANs

CONFIGURATION OF PORTS ON THE SWITCH

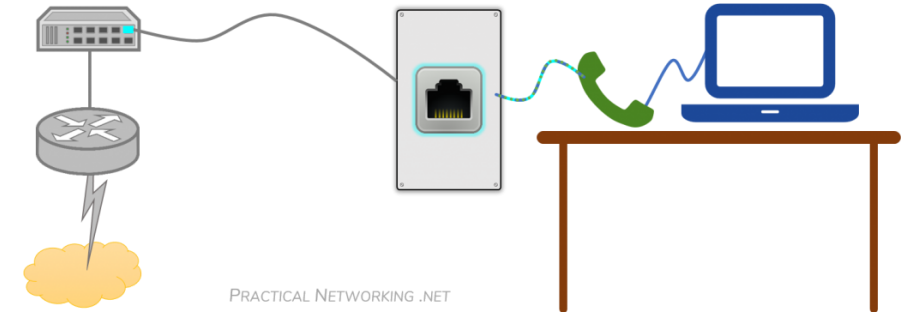
- In typical office environment is IP phone on the desk and PC.
- PC needs to be in separate VLAN as well as IP Phone, so we need 2 separate ports per desk.
- That would mean we can use only half capacity of LAN switch



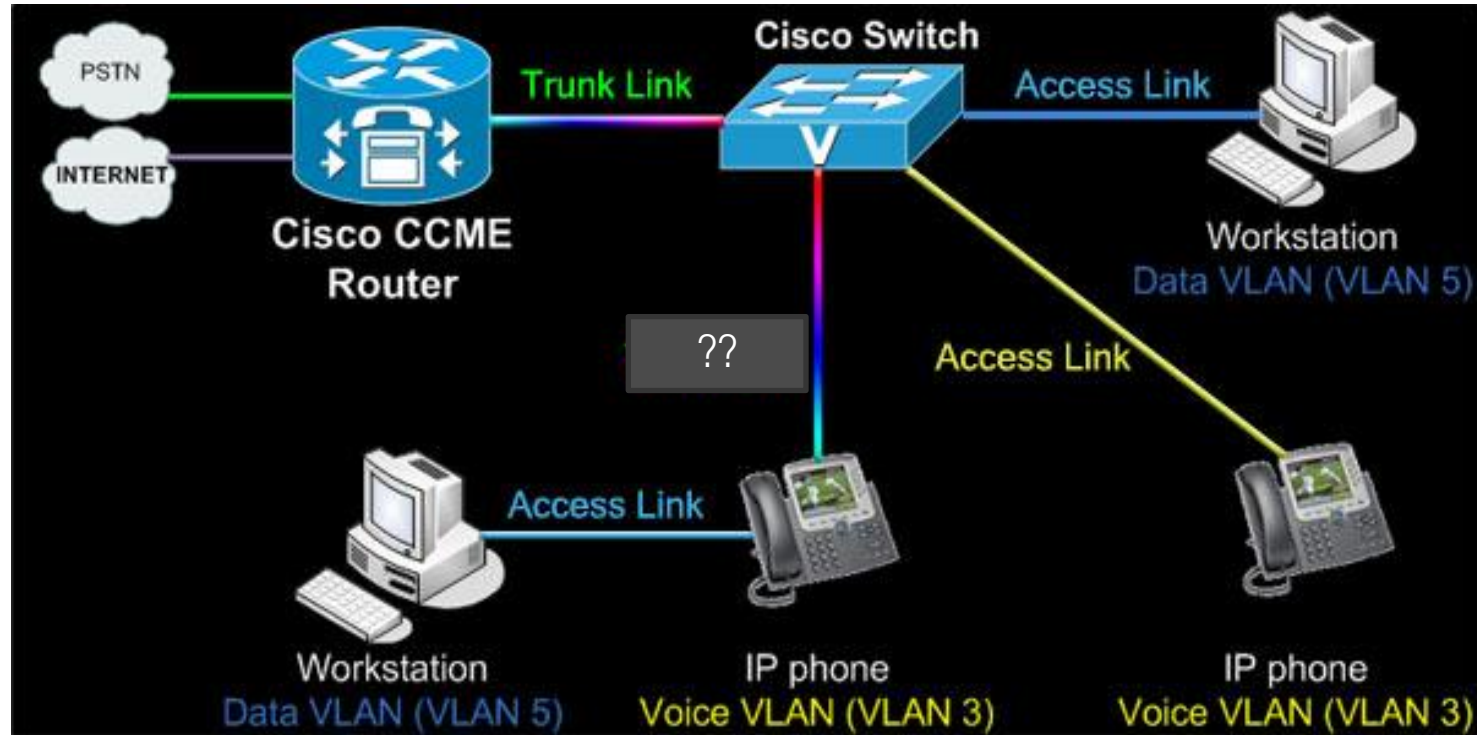
- How to use one port and share data with voice with different VLANs on one port?



Phone has built-in "Switch"



CONFIGURATION OF PORTS ON THE SWITCH

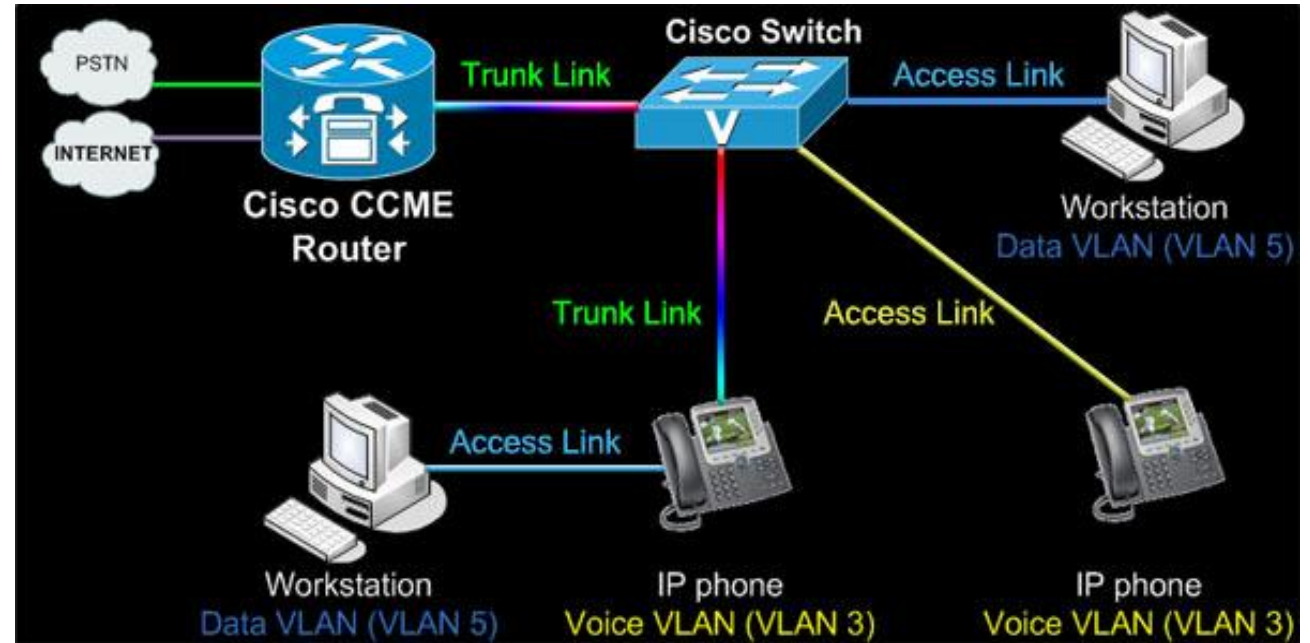


CONFIGURATION OF PORTS ON THE SWITCH

What Is Voice VLAN?

User's interface on the Switch has access mode, so only 1 VLAN is assigned to interface. Cisco invented Auxiliary VLAN (AUX VLAN). The purpose is to allow IP phones to transmit their payloads along with the untagged data coming from a PC.

All the phone traffic is tagged with the ID of the Voice VLAN and the PC traffic is untagged.



Router 1 – Site1

```
hostname Site1
```

```
interface FastEthernet0/0.10
```

```
    encapsulation dot1Q 10
```

```
    ip address 192.168.10.1 255.255.255.0
```

```
interface FastEthernet0/0.50
```

```
    encapsulation dot1Q 50
```

```
    ip address 192.168.50.1 255.255.255.0
```

```
interface FastEthernet0/1
```

```
    ip address 172.16.0.1 255.255.255.252
```

Router 2 – Site2

```
hostname Site2
```

```
interface FastEthernet0/0.10
```

```
    encapsulation dot1Q 10
```

```
    ip address 10.0.10.1 255.255.255.0
```

```
interface FastEthernet0/0.50
```

```
    encapsulation dot1Q 50
```

```
    ip address 10.0.50.1 255.255.255.0
```

```
interface FastEthernet0/1
```

```
    ip address 172.16.0.2 255.255.255.252
```

Router 1 – Site1

ip dhcp pool DATA

network 192.168.10.0 255.255.255.0

default-router 192.168.10.1

ip dhcp pool VOICE

network 192.168.50.0 255.255.255.0

default-router 192.168.50.1

option 150 ip 192.168.50.1

ip route 10.0.0.0 255.255.0.0 172.16.0.2

Router 2 – Site2

ip dhcp pool DATA

network 10.0.10.0 255.255.255.0

default-router 10.0.10.1

ip dhcp pool VOICE

network 10.0.50.0 255.255.255.0

default-router 10.0.50.1

option 150 ip 10.0.50.1

ip route 192.168.0.0 255.255.0.0 172.16.0.1

Switch 1 – Site1

```
hostname Switch1
```

```
interface range fa0/1-9
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
switchport voice vlan 50
```

```
interface range fa0/10
```

```
switchport mode trunk
```

Switch 2 – Site2

```
hostname Switch2
```

```
interface range fa0/1-9
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
switchport voice vlan 50
```

```
interface range fa0/10
```

```
switchport mode trunk
```

CCME SERVICES

You must now configure the Call Manager Express telephony service on RouterA to enable VoIP on your network.

```
RouterA(config)#telephony-service    #Configuring the router for telephony services#  
RouterA(config-telephony)#max-dn XX  #Define the maximum number of directory numbers#  
RouterA(config-telephony)#max-ephones XX    #Define the maximum number of phones#  
RouterA(config-telephony)#ip source-address 192.168.50.1 port 2000    #IP Address source; port 2000 is for SCCP#
```

Router 1 – Site1 (1XXX)	Router 2 – Site2 (2XXX)
telephony-service	telephony-service
max-ephones 10	max-ephones 10
max-dn 144	max-dn 144
ip source-address 192.168.50.1 port 2000	ip source-address 10.0.50.1 port 2000

CCME SERVICES

Although IP Phone is already connected to Switch, it needs additional configuration before being able to communicate.

- **Configure phone numbers which will be used – associated to IP phones**

```
RouterA(config)#ephone-dn 1      #Defining the "first" directory entry#
```

```
RouterA(config-ephone-dn)#number 42851      #Assign the phone number to this entry#
```

- **Configure IP phone based on MAC address - each phone is in config called “ephone”**

```
RouterA(config)#ephone 1      #Select phone based on ephone tag number#
```

```
RouterA(config-ephone)#mac-address XXXX.XXXX.XXXX      #Assign MAC address of Hardware phone to ephone#
```

CCME SERVICES

Now you have to associate number with ephone:

RouterA(config-ephone)#button 1:1 *#Assign to 1st button directory entry 1 (e.g. number 42851)#*

Separator ":" means normal ring.
There can be other, like "b" for
beep; or "s" for silent...



SEARCHING FOR DESTINATION OF CALL

VoIP don't read phone numbers but looks for IP addresses to send a call

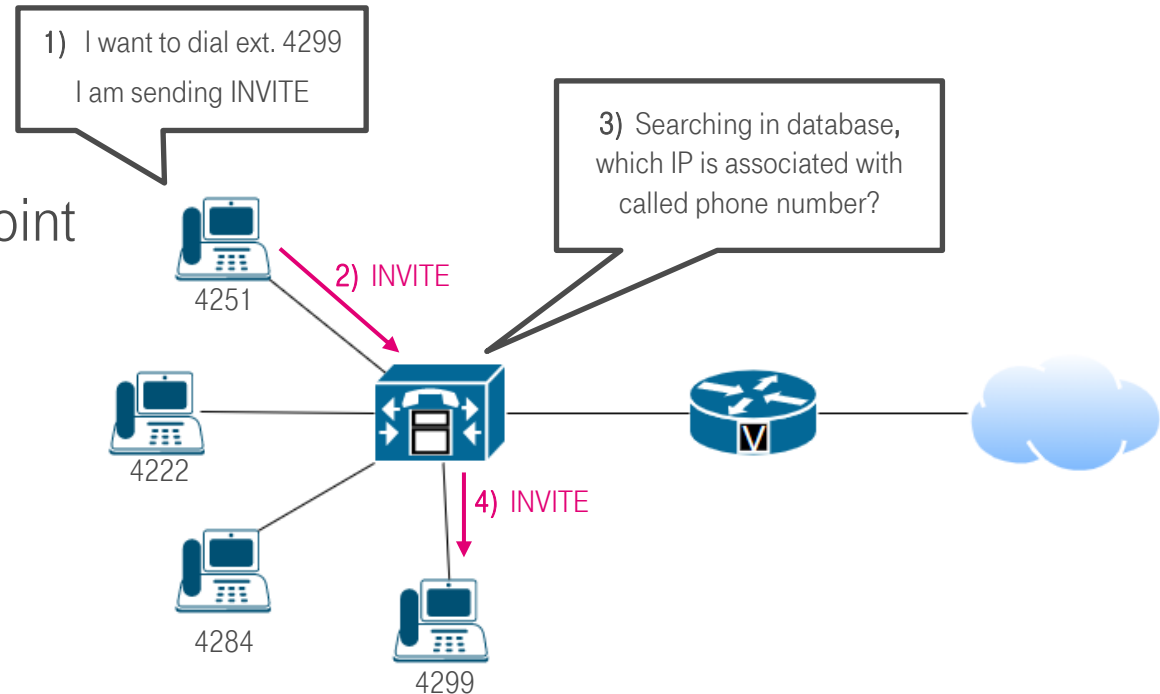
The **central call processor (i.e. CUCM)** is hardware that runs a specialized database/mapping program called a **soft switch**. Sending data is handled by **soft switches**.

Soft switches know:

Where the network's endpoint is (on/off the net)

What phone number is associated with that endpoint

The endpoint's current IP address

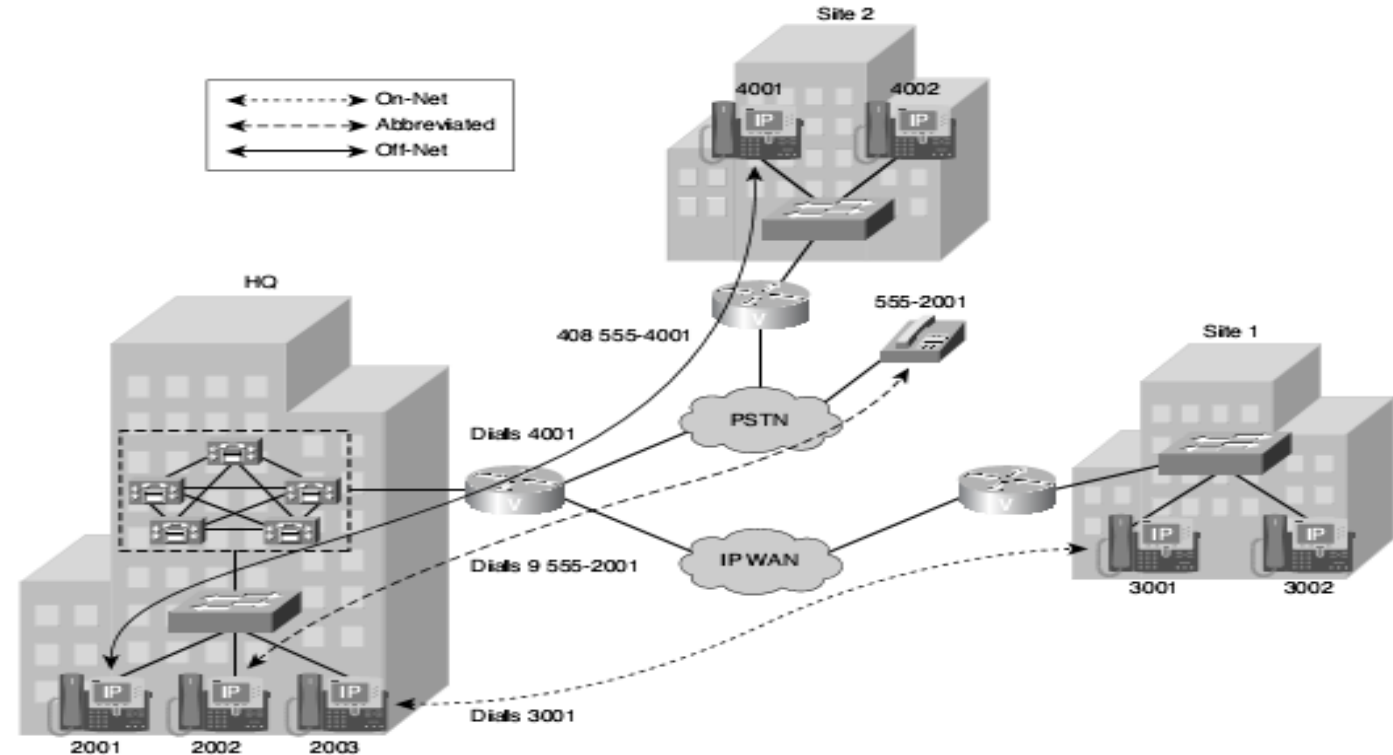


TYPE OF CALLS

On-Net: calls remain within the CUCM system (private network).

Off-Net: calls are normally calls that are routed outside the private telephony system to the PSTN.

Abbreviated dialing: is the idea that employees do not need to dial the full ten-digit phone number.



Router 1 – Site1 (1XXX)

ephone-dn 100

number 1000

ephone-dn 101

number 1001

ephone-dn 104

number 1004

ephone-dn 110

number 1010

ephone-dn 103

number 1003

ephone-dn 102

number 1002

ephone 1

button 1:100

ephone 2

button 1:101

ephone 3

button 1:102

Router 2 – Site2 (2XXX)

ephone-dn 1

number 2001

ephone-dn 2

number 2002

ephone-dn 3

number 2003

ephone-dn 4

number 2004

ephone-dn 5

number 2005

ephone 1

button 1:1

ephone 2

button 1:2

ephone 3

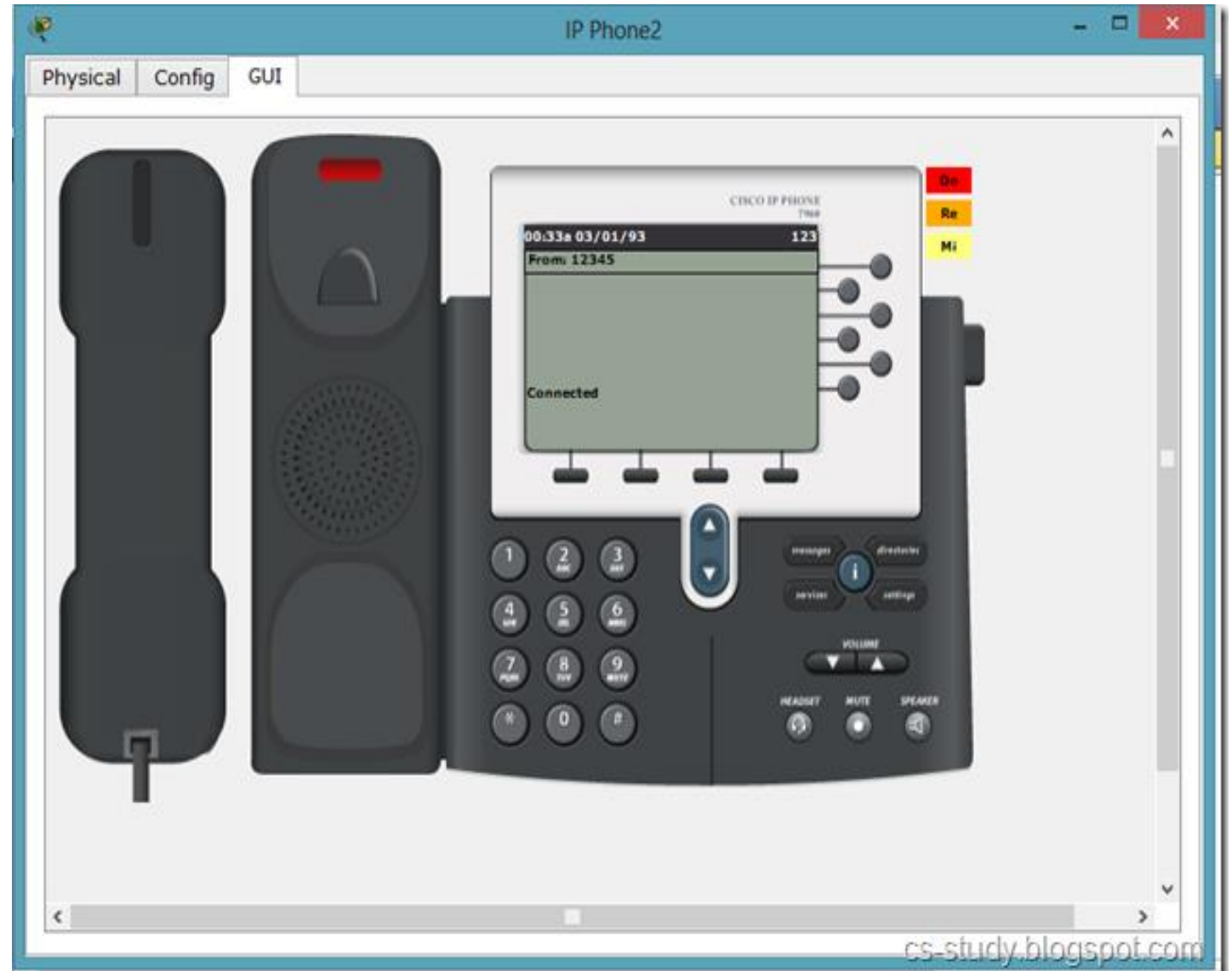
button 1:3

ephone 4

TEST FUNCTIONALITY

Now, we can see that in GUI mode of Phone, we have a phone number available. That means phone is working and communicating with CCME.

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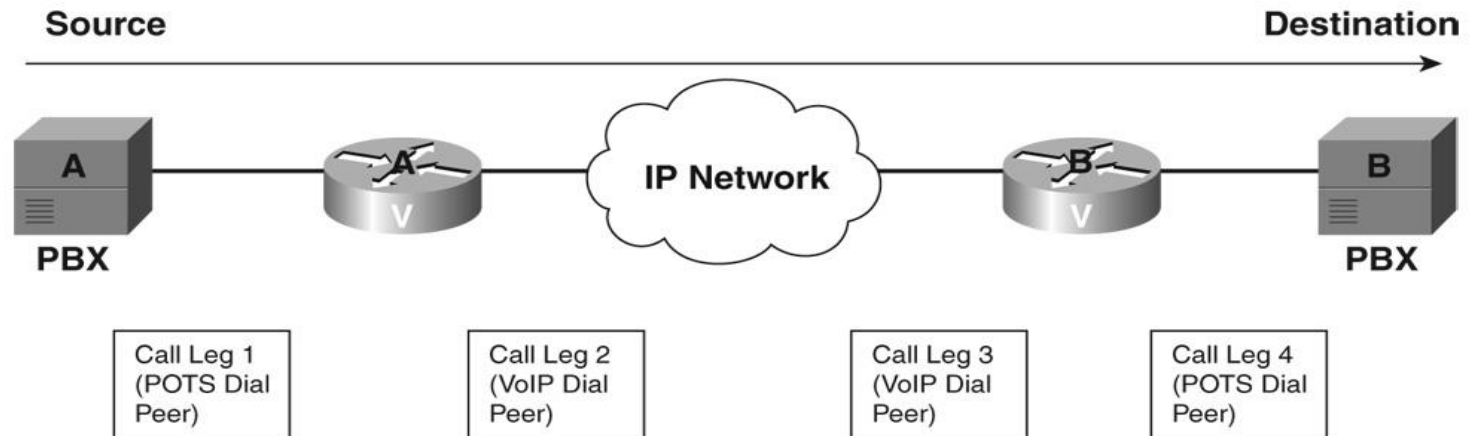


CALL LEG

is a logical connection (segment) between two routers or between a router and a telephony-capable device.

A typical voice call might be composed of four call legs—two from the perspective of the originating router and two from the standpoint of the terminating router. Similar to static routes, dial peers must be configured for each call leg to complete the path.

Example: A phone attached to PBX A places a call destined for a phone attached to PBX B. PBX A forwards the call to Router A using a POTS dial peer. Router A forwards the call to Router B using a VoIP dial peer. Router B forwards the call to PBX B using a POTS dial peer. PBX B connects the call to the destination phone.



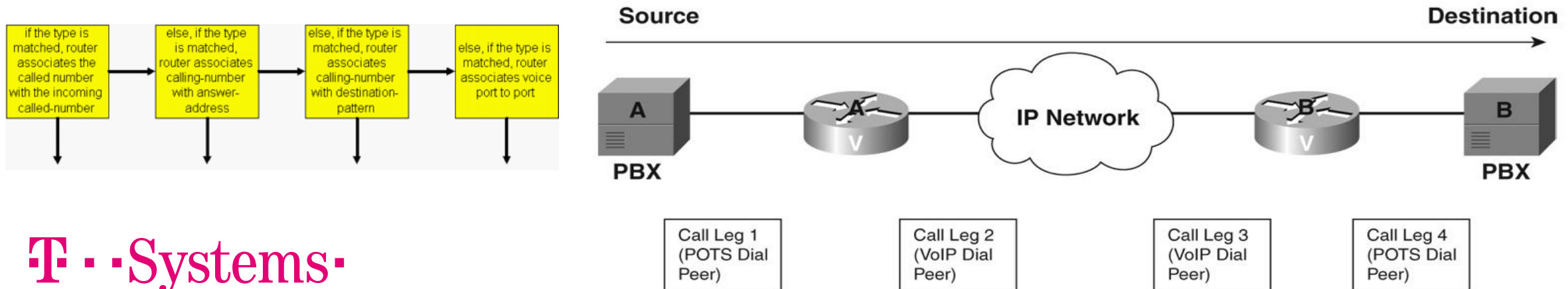
DIAL-PEER

A dial peer, also known as an addressable call endpoint, is a device that can originate or receive a call in a telephone network.

Depending on the call leg, a call is routed using one of the two types of dial peers:

- Voice-network dial peers - include VoIP-capable computers, routers, and gateways within a network.
- POTS dial peers include traditional telephone network devices such as phone sets, cell phones, and fax machines.

Dial peers are used for both inbound and outbound call legs. The router selects a dial peer for a call leg by matching the string that is defined by using the **destination-pattern** (outbound dial peer), **answer-address** (inbound dial peer) or **incoming called-number** (inbound dial peer)



DESTINATION PATTERNS

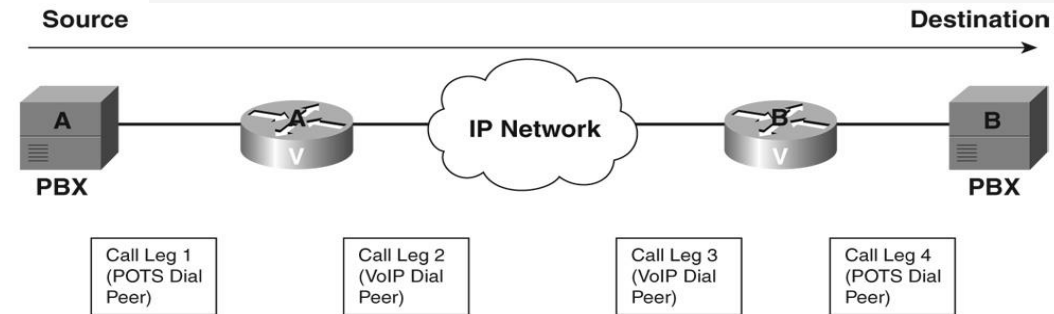
In order to match **outbound** dial peers, the router or gateway uses the dial peer **destination-pattern** *called_number* command.

On POTS dial peers, the port command is then used to forward the call.

On Voice-Network dial peers, the session target command is then used to forward the call.

```
dial-peer voice 3 pots <--POTS dial peer
destination-pattern 5556123
port 2/0/1

dial-peer voice 4 voip <--VoIP dial peer
destination-pattern 5557000
session target ipv4:12.91.0.1
```



PATTERNS USED ON CUCM EXPRESS OR VOICE GW

Symbol Description

- . Indicates a single-digit placeholder. For example, 555.... matches any dialed string beginning with 555, plus at least four additional digits.
- [] Indicates a range of digits. A consecutive range is indicated with a hyphen (-); for example, [5-7]. A nonconsecutive range is indicated with a comma (,); for example, [5,8]. Hyphens and commas can be used in combination; for example, [5-7,9].
Note Only single-digit ranges are supported. For example, [98-102] is invalid.
- () Indicates a pattern; for example, 408(555). It is used in conjunction with the symbol ?, %, or +.
- ? Indicates that the preceding digit occurred zero or one time.
- % Indicates that the preceding digit occurred zero or more times.
- + Indicates that the preceding digit occurred one or more times.
- T Indicates the interdigit timeout. The router pauses to collect additional dialed digits.

WILDCARD EXAMPLE

Using Dial-Peer Wildcards:

- Period (.): Will match any digit
- Plus(+): matches one or more instances of the preceding digits
- Brackets ([]): Matches a range of digits
- T: matches any dialed number from 0-32 digits
- Carrot (^): Does not match
- Comma (,): Inserts a one-second pause between dialed digits

- Example:

- | | |
|-----------------|----------|
| ✓ 555[1-3]... | Matches: |
| ✓ 5+23 | Matches |
| ✓ [14-6]555 | Matches |
| ✓ 55[59]12 | Matches |
| ✓ [^1-7]..[135] | Matches |

WILDCARD EXAMPLE

Using Dial-Peer Wildcards:

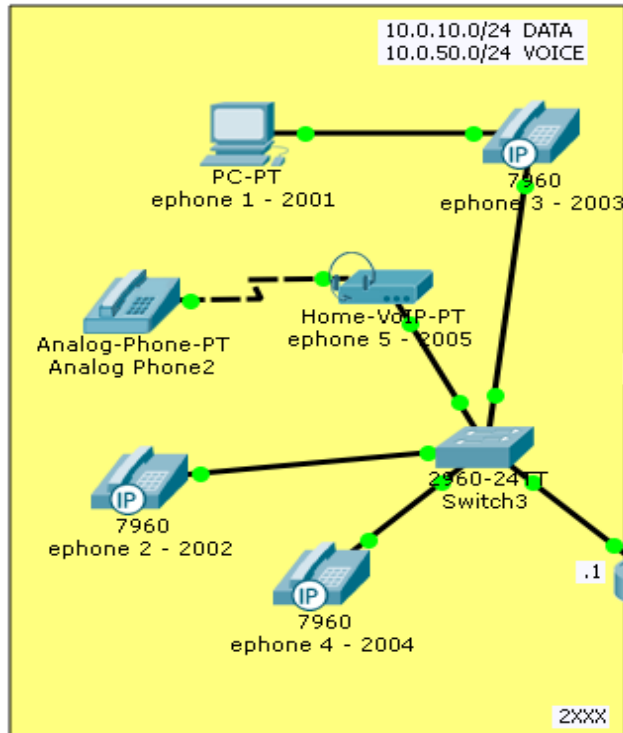
- Period (.): Will match any digit
- Plus(+): matches one or more instances of the preceding digits
- Brackets ([]): Matches a range of digits
- T: matches any dialed number from 0-32 digits
- Carrot (^): Does not match
- Comma (,): Inserts a one-second pause between dialed digits
- Example:
 - ✓ 555[1-3]... Matches: 5551..., 5552..., 5553... (Where ... is any three digits)
 - ✓ 5+23 Matches 5523, 55523, 55523
 - ✓ [14-6]555 Matches 1555, 4555, 5555, 6555
 - ✓ 55[59]12 Matches 55512, 55912
 - ✓ [^1-7]..[135] Matches 8..1, 8..3, 8..5, 9..1, 9..3, 9..5 (Where ^ is any two digits)

• Router 2 – Site2

dial-peer voice 1000 voip

destination-pattern 1...

session target ipv4:172.16.0.1

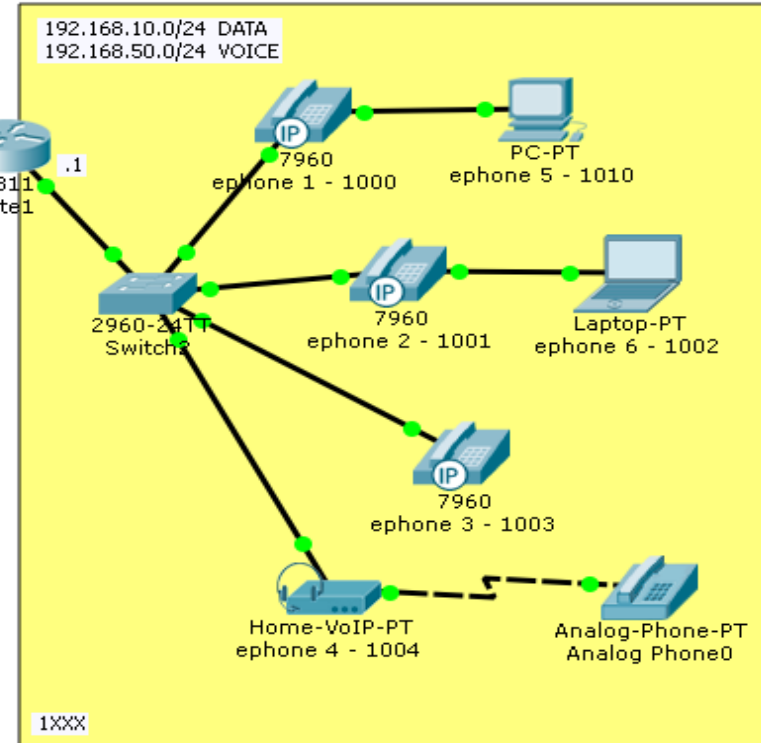


• Router 1 – Site1

dial-peer voice 2000 voip

destination-pattern 2...

session target ipv4:172.16.0.2



DIAL PLAN

is the way in which the PSTN carrier networks know how to perform its call routing.

Directory Number Range	Usage	DID Ranges	Non-DID Ranges
0XXX	Excluded: 0 is used to reach the operator.	—	—
1XXX	Site A extensions	418 555 1XXX	Not applicable
2XXX	Site B extensions	919 555 2XXX	Not applicable

Direct inward dial (DID) = direct full number

International DID numbers can be purchased from international providers.

The format of telephone numbers is standardized by ITU-T in the recommendation E.164.

THE NUMBERING PLAN E.164 (DID)

E.164 recommendation, also called the "international public telecommunications numbering plan," was published by the ITU-T:

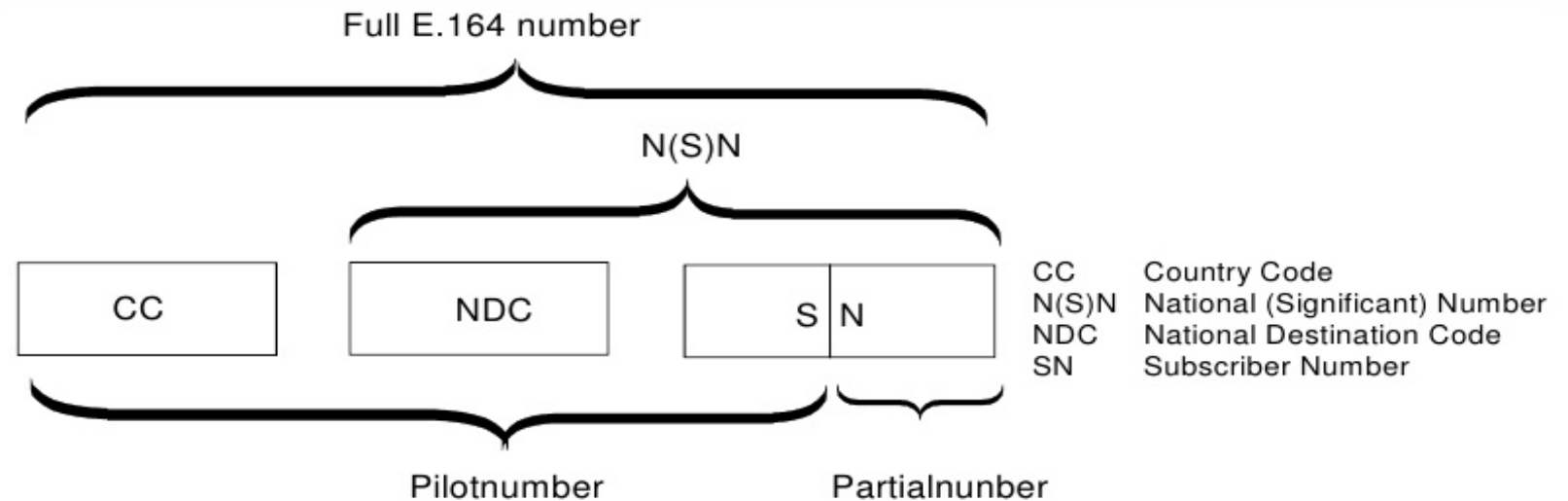
A telephone number can have a maximum of 15 digits

The first part of the telephone number is the country code (one to three digits)

The second part is the national destination code (NDC)

The last part is the subscriber number (SN)

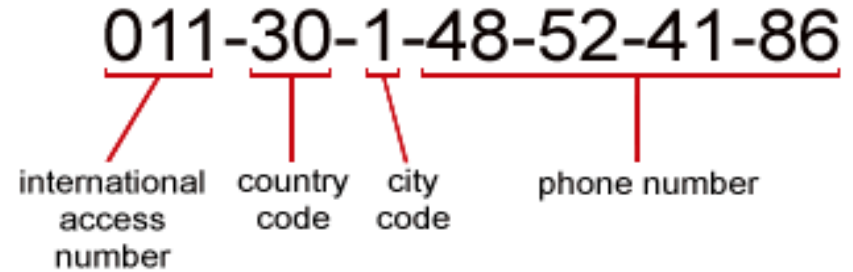
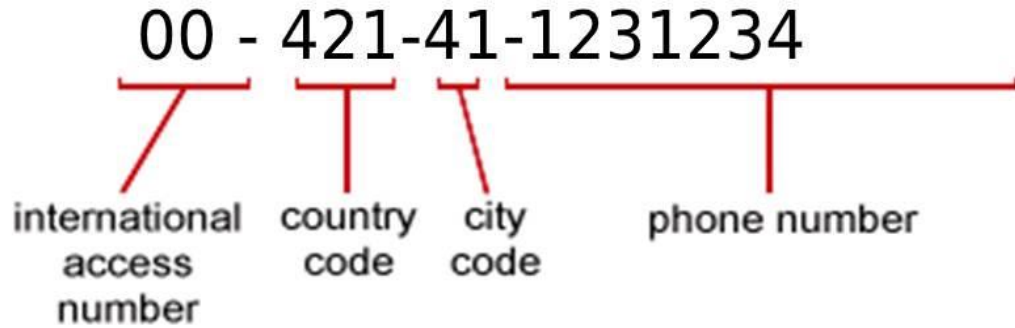
The NDC and SN together are collectively called the national (significant) number



MAKING INTERNATIONAL CALL

routed through your long-distance phone carrier to another country's long-distance phone carrier

you have to dial two separate numbers, your country's **exit code** (or **international access code**) and the corresponding **country code** of the place you're calling. Usually **exit codes** are either 00 or 011 (some exceptions existing)



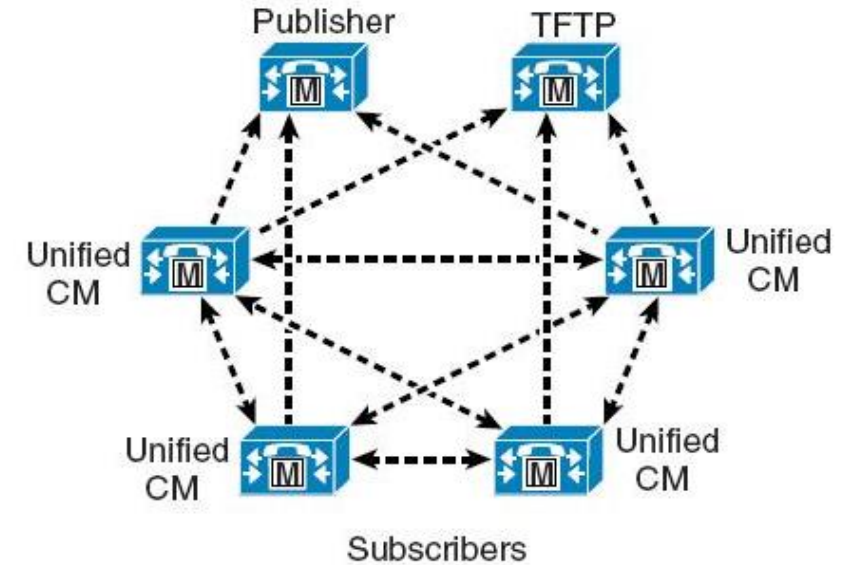
CISCO UNIFIED COMMUNICATIONS



T · · Systems ·

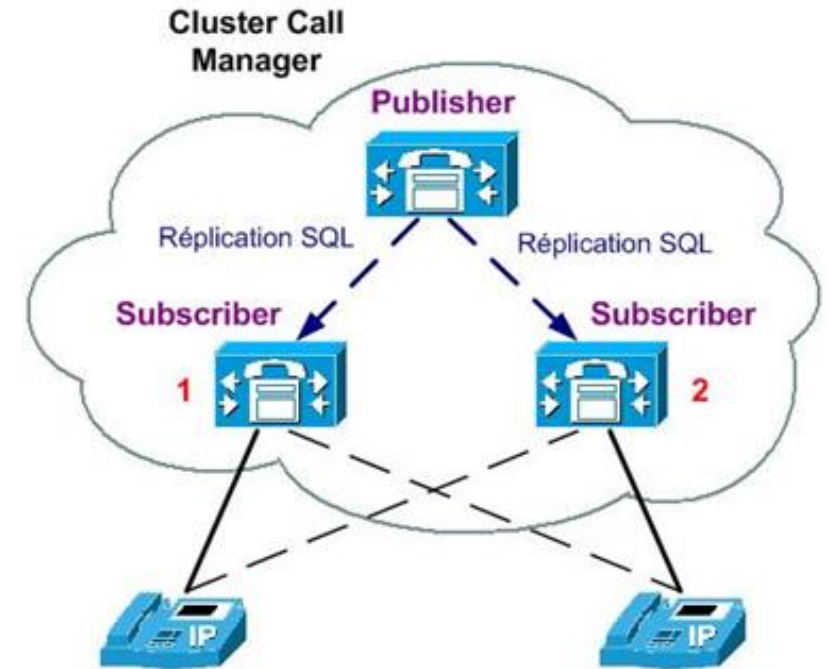
WHAT IS CUCM

- is the call-routing component of the Cisco UC ecosystem, providing call setup and teardown services to both voice and video communications.
- CUCM is a software solution, running operating system based on Linux. OS and application updates are provided by Cisco.
- CUCM supports clustering of servers to provide high availability and scalability. Database redundancy is provided by sharing a common database.
- CUCM leverages the IBM Informix Dynamic Server (IDS) to store all configuration data, including the user database.
- Unsupported software and applications cannot be uploaded or installed into the system



WHAT IS CUCM

- CUCM cluster can have up to 20 servers in it.
 - 1 publisher server - maintains the read/write copy of the CUCM's database.
 - 4 pairs of subscriber servers - publisher replicates the database as a read-only database to subscriber servers in the CUCM cluster.
 - The additional 11 servers in the cluster are responsible for various services, including TFTP and media resources (conferencing, music on hold, transcoding, ...) and integration with third-party applications (CTI servers, ...)



Each phone communicates with 1 CUCM Subscriber. If connection fails, phone connects to backup

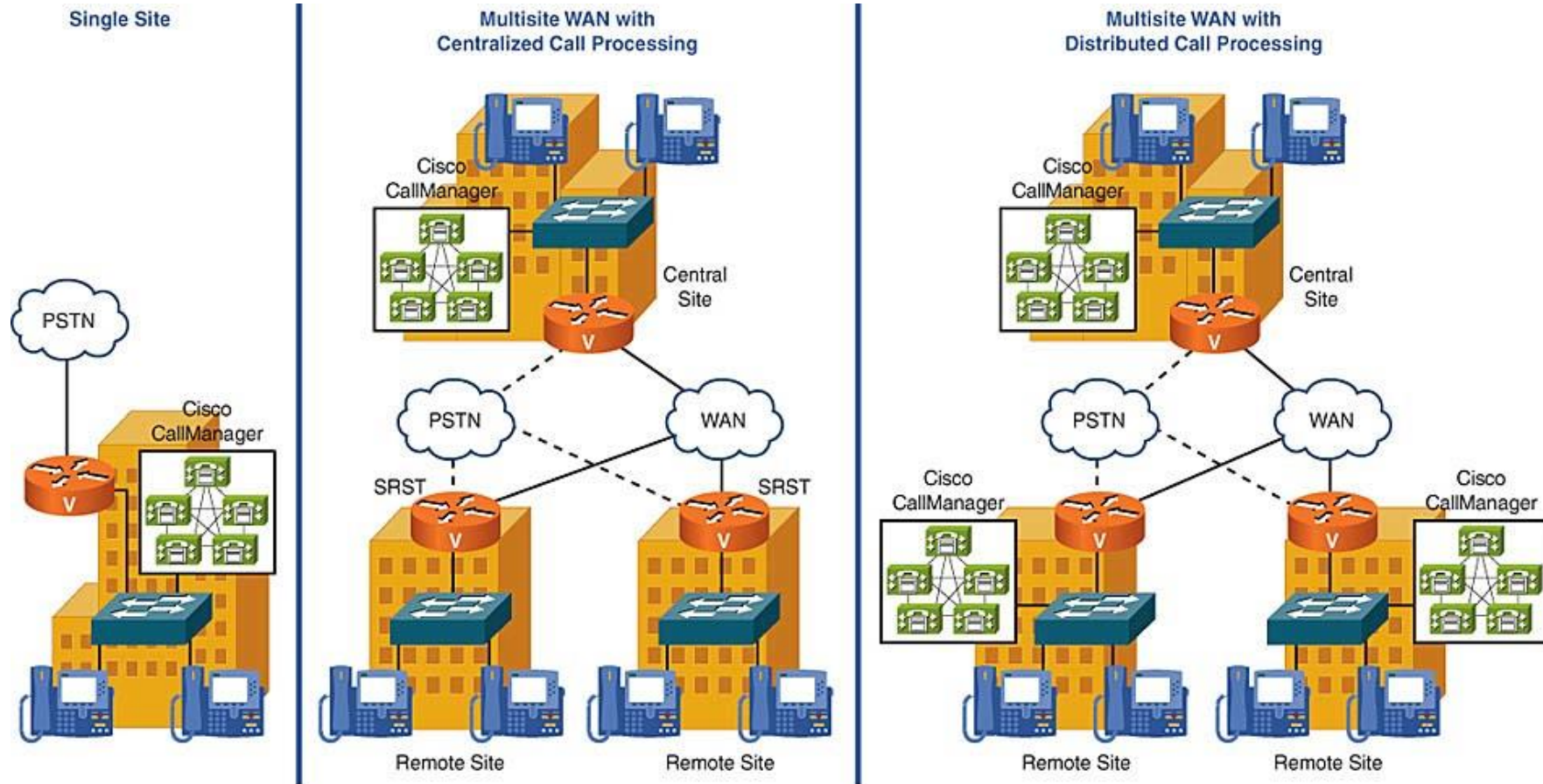
CUCM FUNCTIONS

CUCM extends enterprise telephony features and functions to packet telephony network devices

It provides these functions:

- Call processing - routing and terminating calls, including billing and statistical collection processes
- Signaling and device control - coordinates all signaling events between call endpoints
- Dial plan administration - responsible for digit analysis of all calls
- Phone feature administration - services such as hold, transfer, forward, conference, speed dial, redial, and call park
- Directory services - CUCM uses a portion of the Informix Database Server (IDS), Lightweight Directory Access Protocol version 3 (LDAPv3) database to store user information
- Backup and restore tools

MODELS OF CUCM CLUSTER



- CUCM cluster can handle up to 30000+ phones (7500 each SUB)
- 100+ sites

SURVIVABLE REMOTE SITE TELEPHONY

SRST is a Cisco Unified Communications Manager (CUCM) call processing backup mechanism that allows Cisco IP phones to register to a Cisco router. A Cisco router in SRST mode provides call setup and call teardown services similar to CUCM, but only when connectivity to the CUCM servers provided in the Cisco Call Manager group are not reachable.

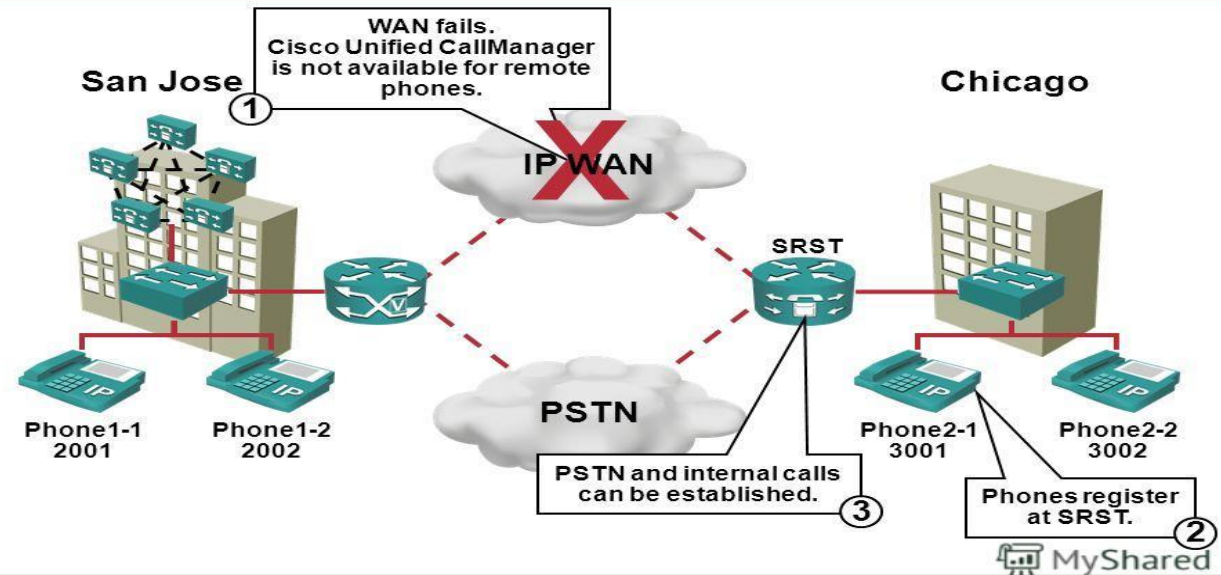
If WAN connectivity is lost between CUCM and the branch site, Cisco IP phones will re-register to their SRST reference provisioned in the device pool configuration.

Only H.323 can be used for SRST gateway

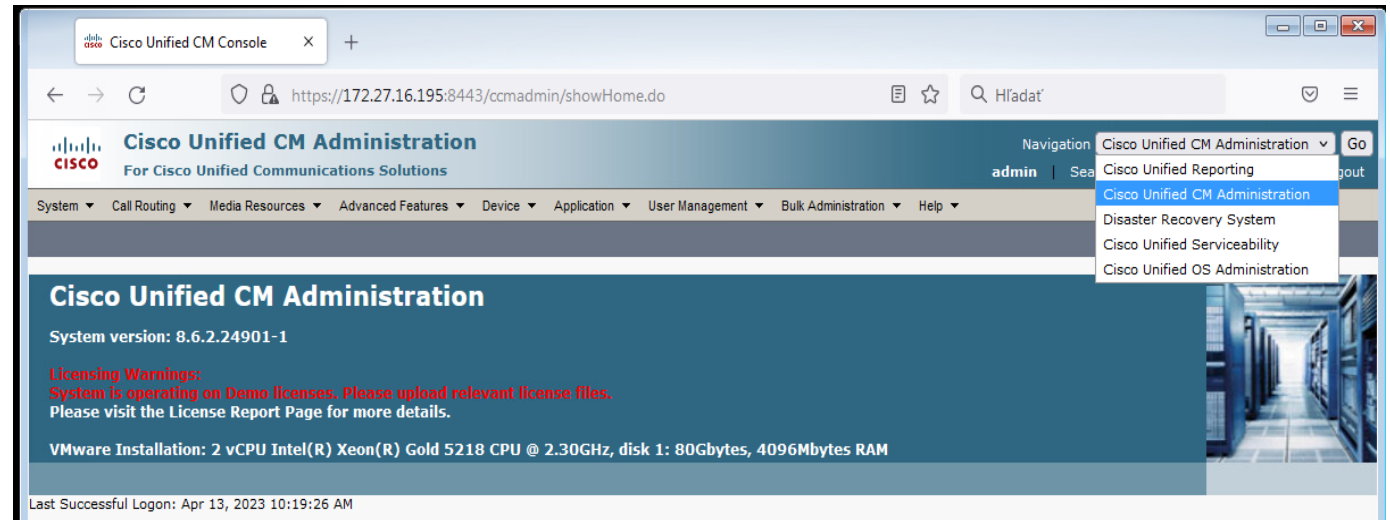
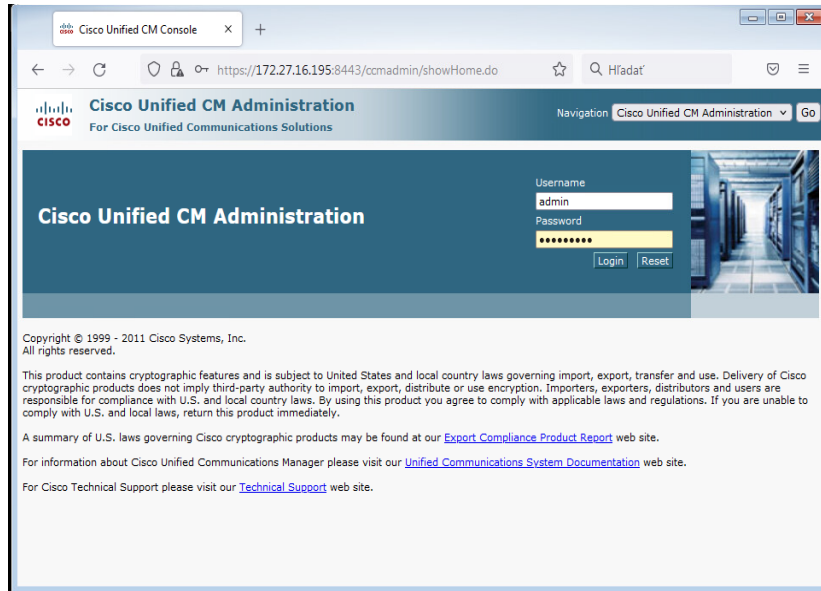
T · · Systems ·

Cisco IP phones send a TCP keepalive to their primary CUCM server every 30 seconds by default. If 3 keepalives are missed, the Cisco IP phone will re-register with another CUCM server in their Call Manager group configuration and attempt to register with the SRST reference if no CUCM servers are available.

Example: SRST Functionality



WELCOME TO CUCM 8.6 DEMO (VERY VERY OLD – EOL 😊)



LAB: ADD NEW PHONE

In CUCM administration: Device -> Phone; Add New

For our purposes choose Cisco IP communicator and SCCP protocol, check and fill out all mandatory fields marked (*) - For device name put in your MAC address in form: SEP <MAC address>

Device Name*

SEP18DBF2150C7B

Confirm settings by Save button and check if IP communicator is registered:

Registration

Registered with Cisco Unified Communications Manager CUCM-LAB-PUB

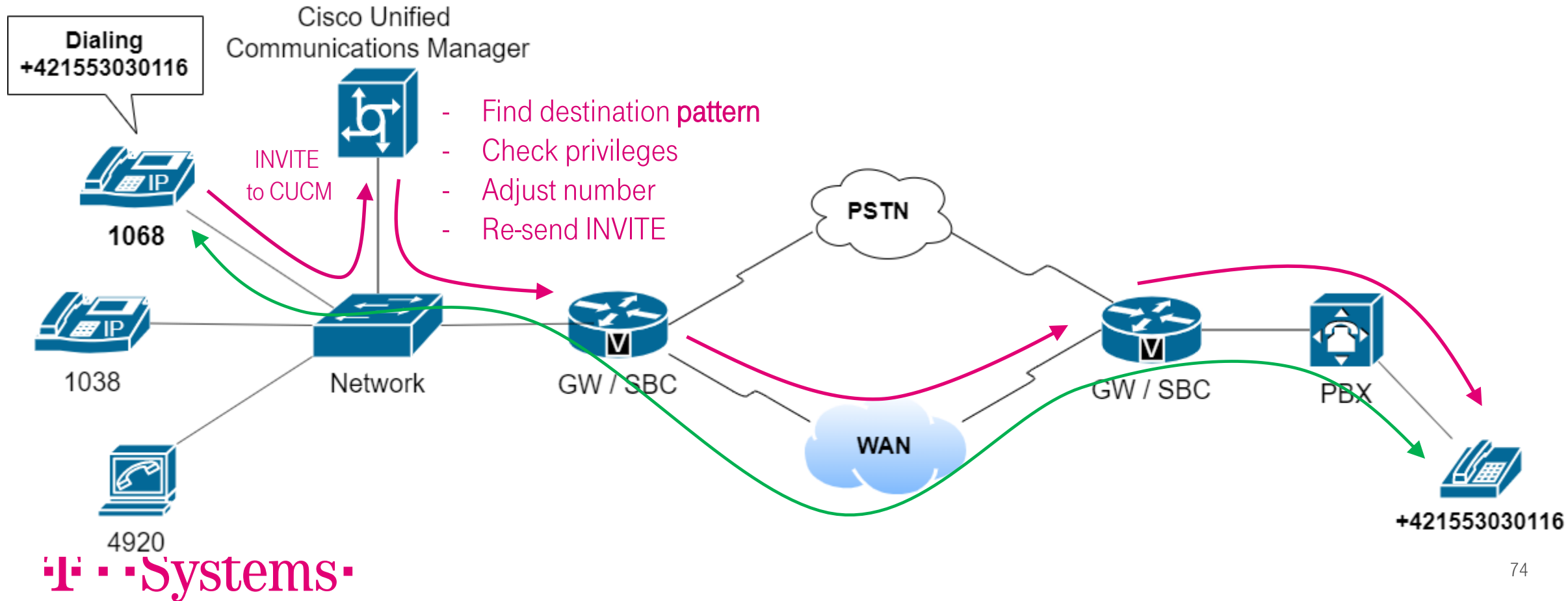
IP Address

[172.27.16.114](#)

Active Load ID

CIPC-8-6-5-0

SIMPLE OFF-NET CALL WITH CISCO



PATTERNS

- CUCM is using patterns for digit manipulation (translation pattern) or find destination (route pattern)

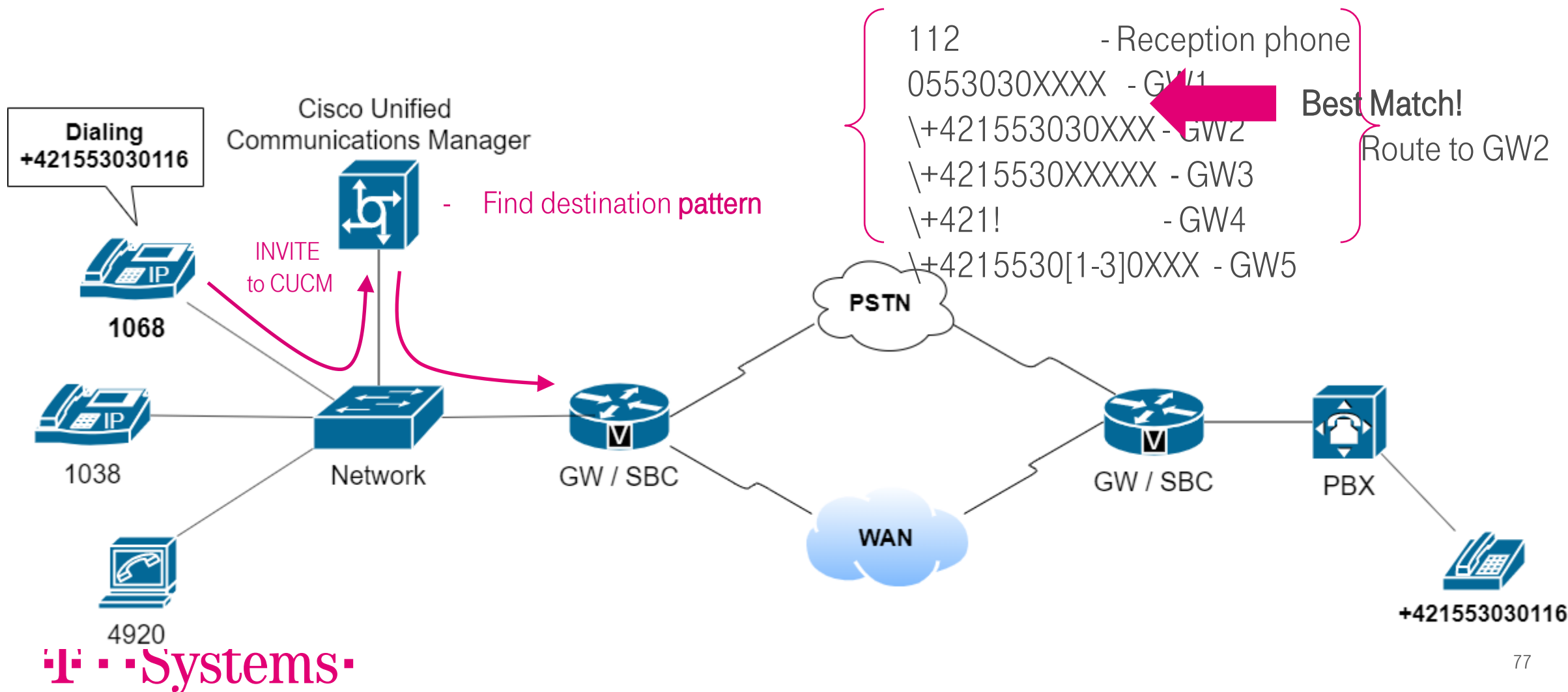
Wildcard	Description
X	Single digit (0–9, *, #)
@	North American Numbering Plan
!	One or more digits (0–9, *, #)
[x–y]	Generic range notation
[^x–y]	Exclusion range notation
.	Terminates access code
#	Terminates interdigit timeout
<wildcard>?	Matches zero or more occurrences of any digit that matches the previous wildcard
<wildcard>+	Matches one or more occurrences of any digit that matches the previous wildcard
\+	Matches the + sign as part of a number. (The \ denotes a delimiter, telling CUCM to process the + as its own character +, rather than the wildcard as previously shown.) The + is used for globalized E.164 call routing.

PATTERNS

- CUCM is using patterns for digit manipulation (translation pattern) or find destination (route pattern)

Route Pattern	Result
1234	One exact dialed digit: 1234.
12XX	Matches dialed digits 1200 to 1299, 12**, 12*#, 12#*, and 12##.
13[25–8]6	Matches dialed digits 1326, 1356, 1366, 1376, and 1386.
13[13–59]X	Matches dialed digits 1310–1319, 1330–9, 1340–9, 1350–9, or 1390–9. The * and # have been excluded for brevity.
13[^3–9]6	Matches dialed digits 1306, 1316, 1326, 13*6, or 13#6.
13!	Matches the string 13 followed by one or more digits.
13!#	Matches the string 13 followed by one or more digits and then ended with a #.
\+!	Matches a phone numbers staring with + and is followed by one or more digits, as used by E.164 numbers.

SIMPLE OFF-NET CALL WITH CISCO



PARTITION

A group of numbers with similar reachability characteristics.

Every phone number/pattern in CUCM can be applied to a partition.

Only devices that are configured with a CSS that has access to the partition will be allowed to call the number.

All phone numbers are in the null partition (Partition <None>) by default. All devices have access to the null partition.

You can associate partitions with a time schedule and a time zone. A time period comprises a start time and end time. A time schedule comprises a group of defined time periods that the administrator associates.

CSS

A device can only call patterns that are in partitions accessible by the calling device. The calling party (device) must have a CSS that includes the partition of the called pattern.

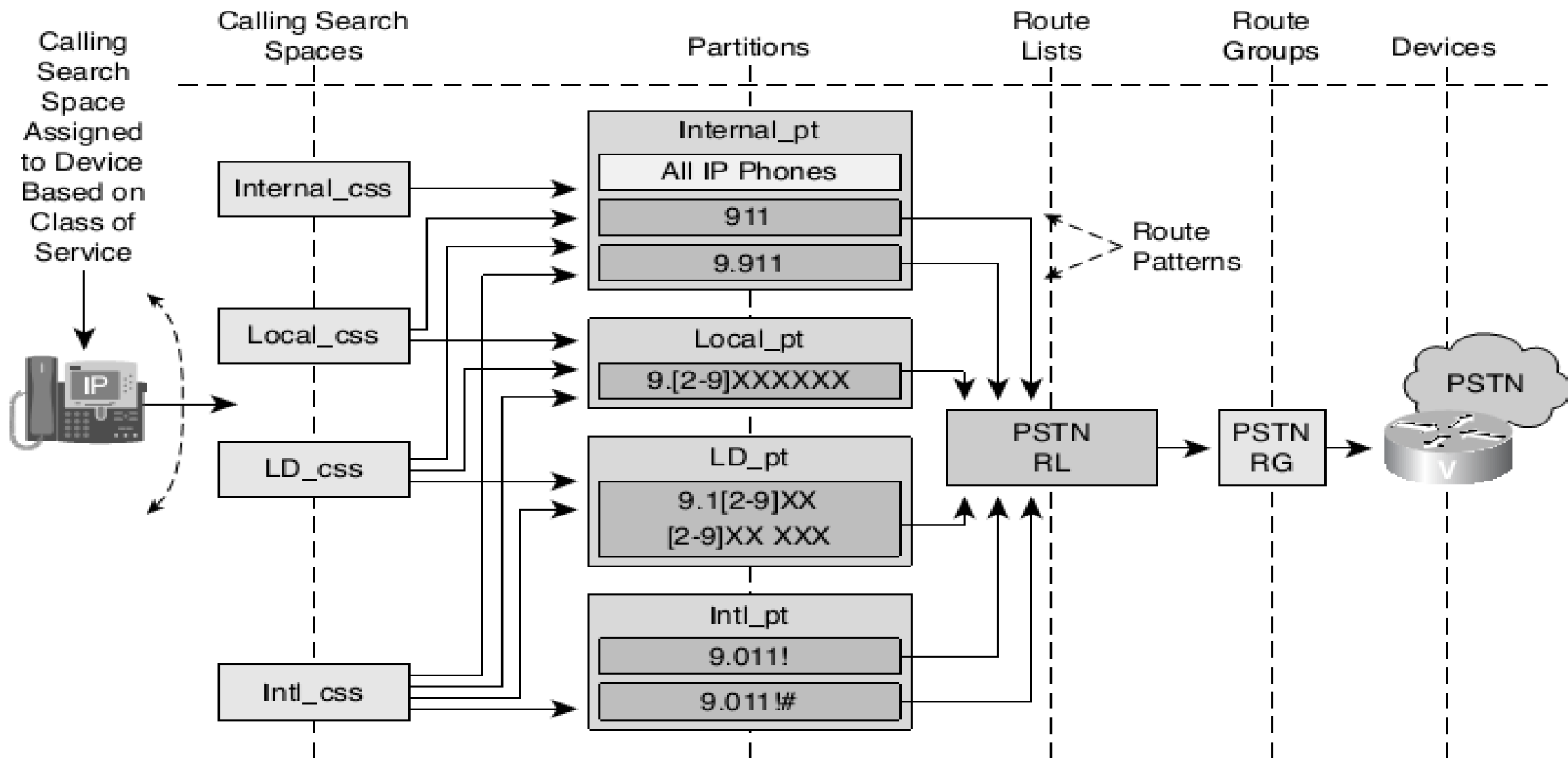
Partitions can be thought of as logical locks.

A CSS is like a key ring that includes multiple keys (partitions).

Partition = where you are
CSS = Where you can call

Partitions:	Lobby_PT	Employee_PT	Manager_PT	No Partition Assigned	Manager_PT
Phones	 Phone 1	 Phone 2	 Phone 3	 Phone 4	 Phone 5
CSSs:	Lobby_PT Employee_PT	Lobby_PT Employee_PT Manager_PT	Lobby_PT Employee_PT Manager_PT Executive_PT	Lobby_PT	No CSS Assigned

EXAMPLE OF SINGLE SITE WITH CSSS

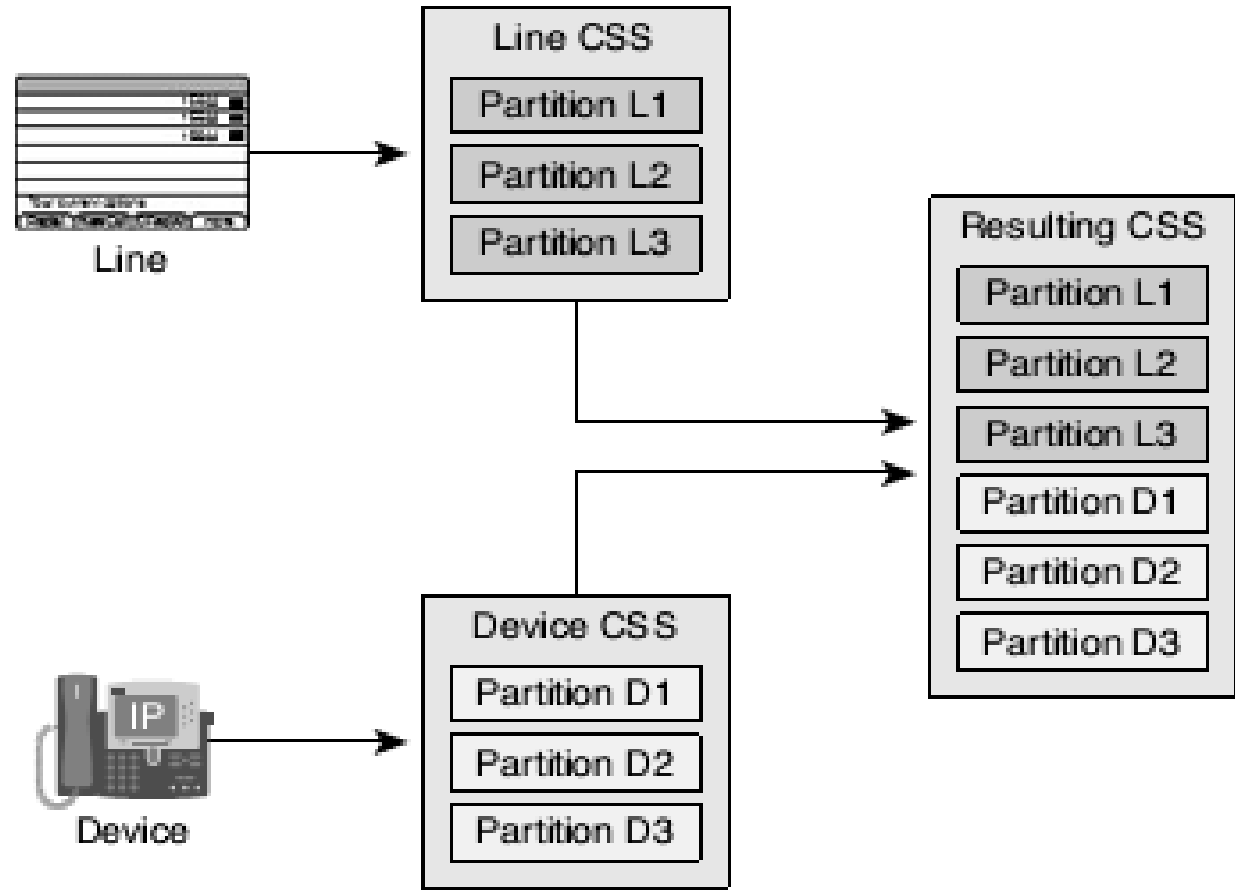


CSS ORDER

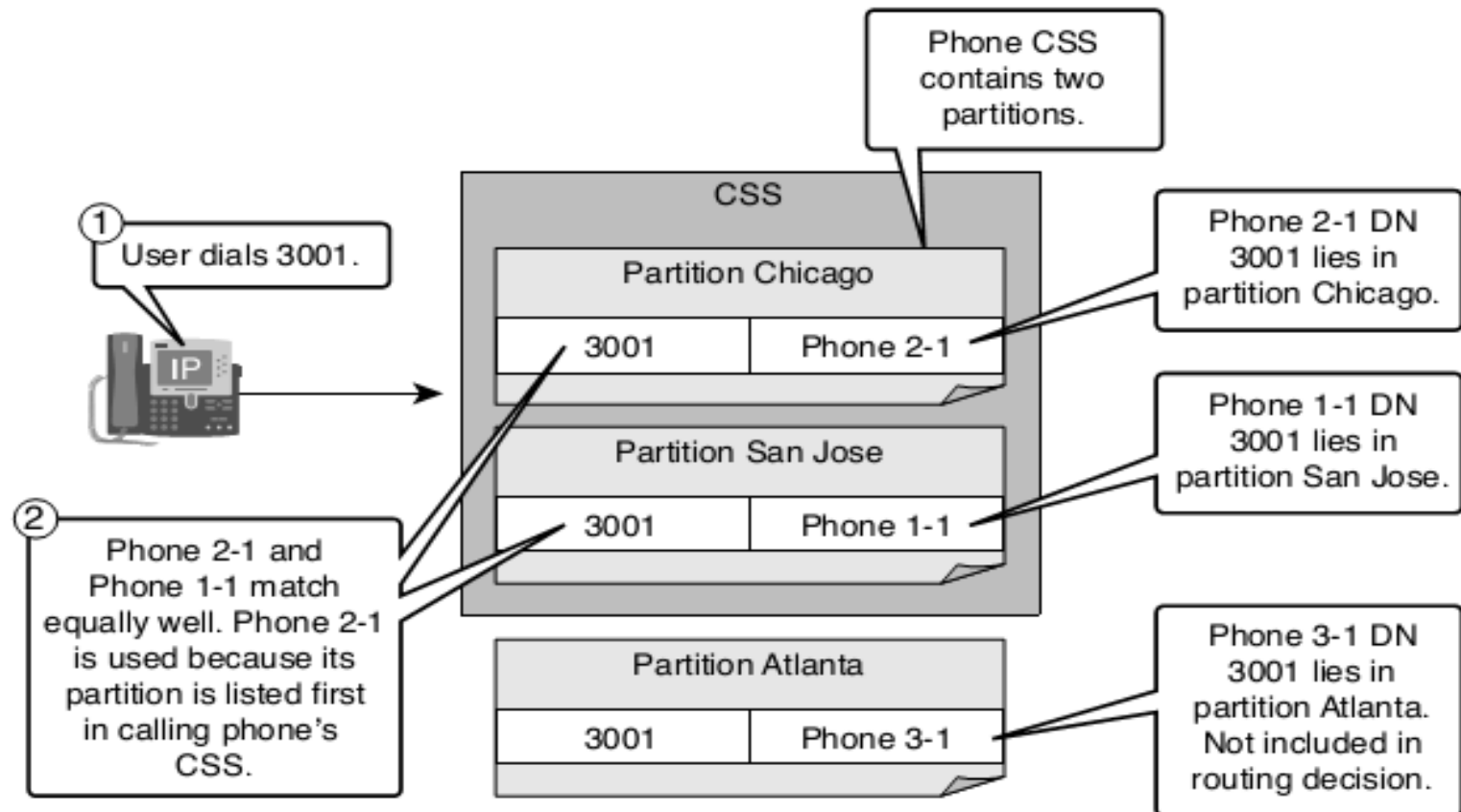
CSS is processed in a top-down manner, but the line CSS will be analyzed before the device CSS.

If a call is explicitly permitted or rejected at the line level, the device CSS will not be considered.

On CTI ports, the line and device CSSs are processed in reverse order; the partitions of the device CSS are processed before the partitions of the line CSS.



A CSS is an ordered list of partitions. The CSS is processed in a top-down fashion in CUCM, which will prioritize partitions in the order in which they are listed out.



DIAL PLAN

Each call routed by CUCM is categorized as one of two types: On-Net and Off-Net.

On-Net: calls remain within the CUCM system (private network).

Off-Net: calls are normally calls that are routed outside the private telephony system to the PSTN.

For Off-net calls you have to configure Route Pattern

ROUTE PATTERN

Used to route calls to remote destinations (PSTN, remote CUCM cluster, ...).

CCM uses route patterns to route or block both internal and external calls.

Wildcard	Description
X	Single digit (0–9, *, #)
@	North American Numbering Plan
!	One or more digits (0–9, *, #)
[x–y]	Generic range notation
[^x–y]	Exclusion range notation
.	Terminates access code
#	Terminates interdigit timeout

@ is a special macro function that expands into a series of patterns representing the entire national numbering plan for a certain country. A route pattern of 9.@ will load a complex 166-route-pattern NANP by default.

! wildcard represents up to 15 digits and is also used for deployments in countries in which dialed digits can be of varying lengths.

NANP DIAL PLAN EXAMPLE

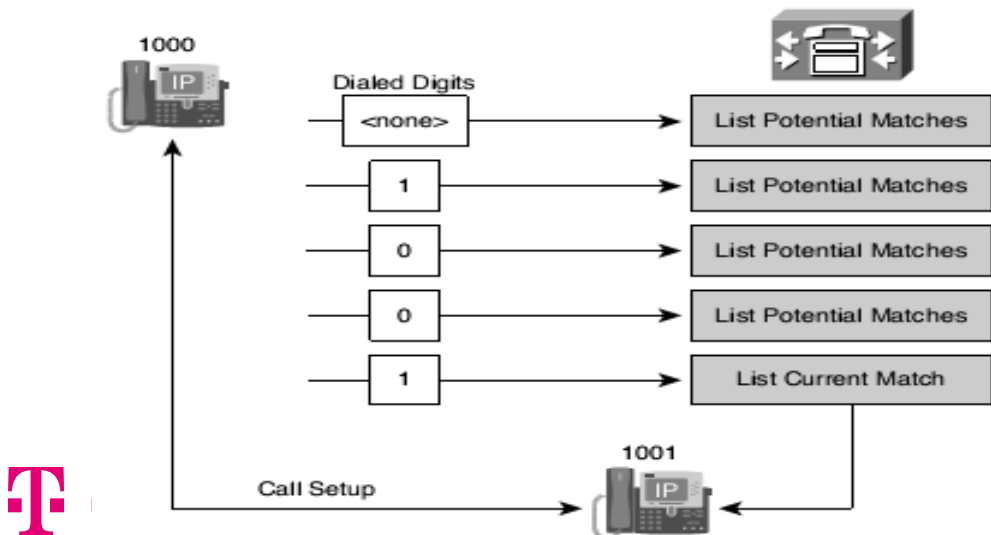
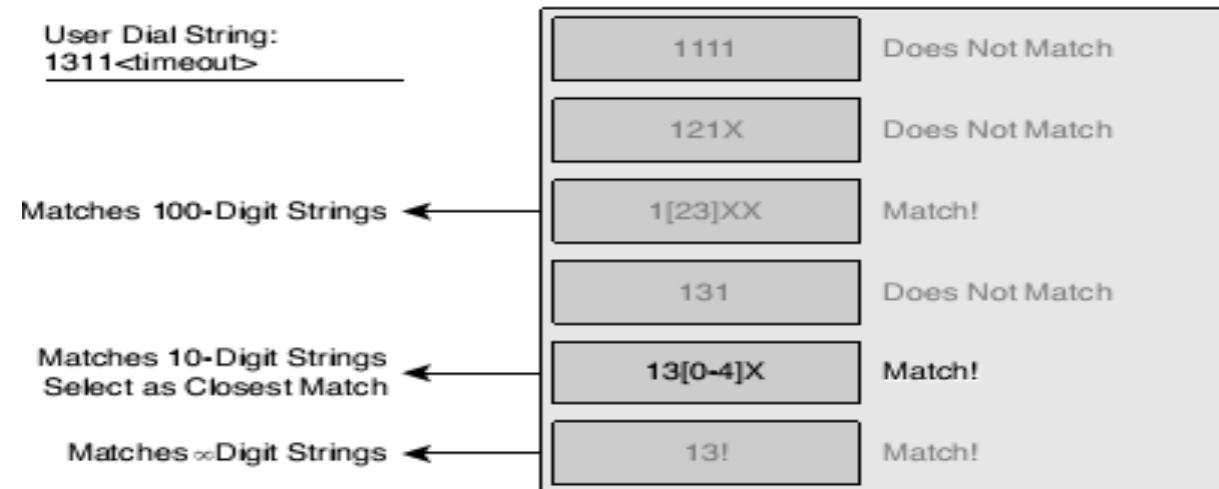
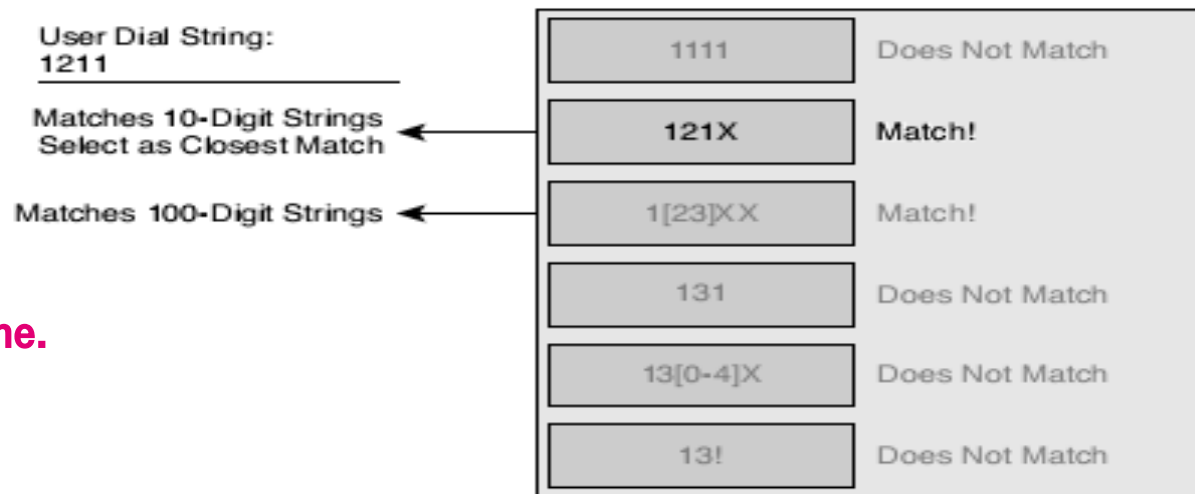
Route Pattern	Description
911	Emergency call routing without access code
9.911	Emergency call routing with access code
9.[2-8]11	Three-digit service codes (for example, 411 for information)
9.[2-9]XX XXXX	Seven-digit local dialing
9.[2-9]XX [2-9]XX XXXX	Ten-digit local dialing
9.1[2-9]XX [2-9]XX XXXX	11-digit long-distance dialing
9.011!	International dialing (variable length)
9.011!#	International dialing (variable length with interdigit timeout termination code [#])

DIGIT ANALYSIS

When an endpoint goes off-hook, CUCM begins the digit analysis process immediately.

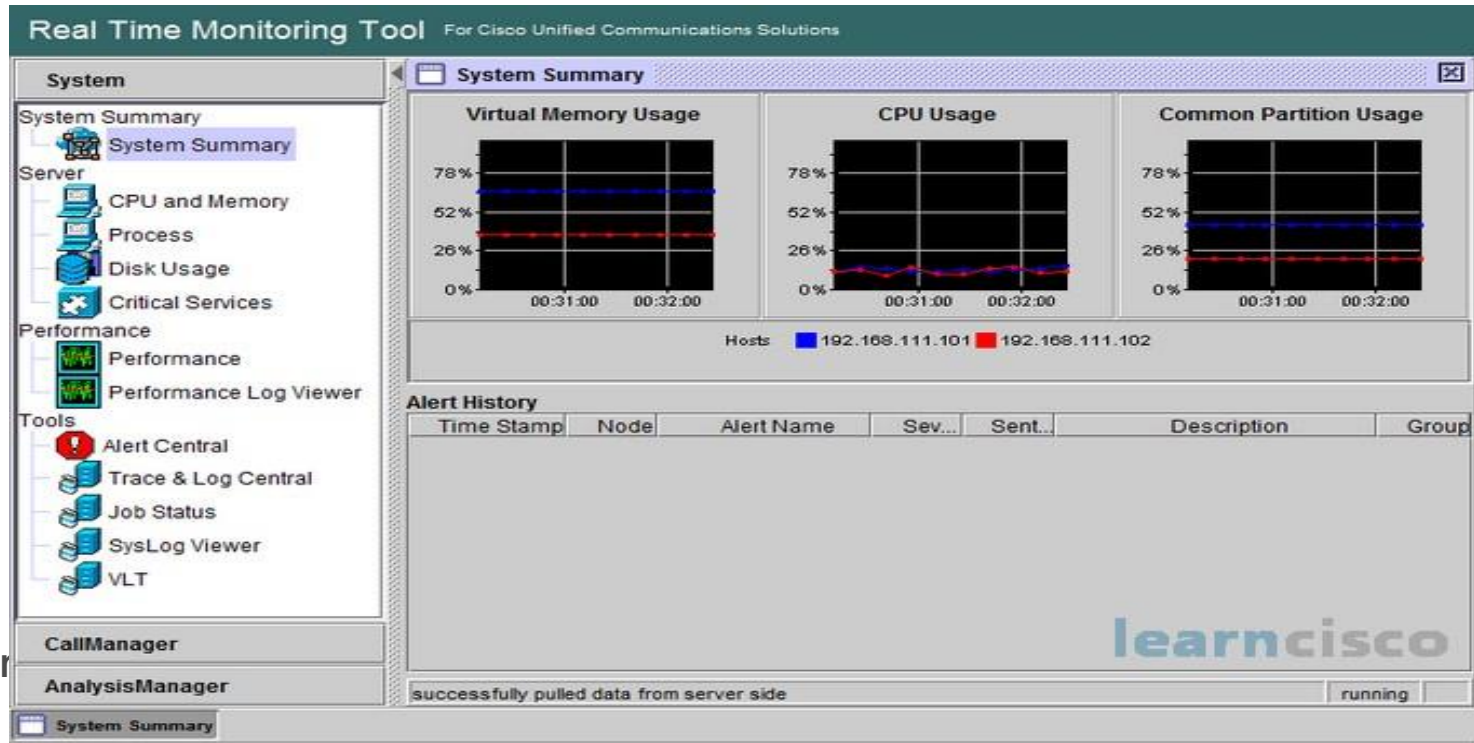
Route pattern provides the closest match for routing.

If two route patterns exactly match the same number of addresses in a partition, the CUCM arbitrarily chooses one.



MONITORING OF CUCM

RTMT available in CUCM as plugin



External tools: Programm